

Optimal Timing of Sequential Distribution: The Impact of Congestion Externalities and Day-and-Date Strategies

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Abstract

The window between a film's theatrical and video releases has been steadily declining with some studios now testing day-and-date strategies (i.e., when a film is released across multiple channels at once). We present a model of consumer choice that examines trade-offs between substitutable products (theatrical and video forms), the possibility of purchasing both alternatives, and a congestion externality affecting consumption at theaters; this permits a normative study of the impact of smaller release windows (0-3 months) for which there is a scarcity of relevant data. We characterize the market conditions under which a studio should pursue direct-to-video, day-and-date, and delayed video release strategies. During seasons of peak congestion, we establish that day-and-date strategies are optimal for films with high content durability (i.e., films whose content tends to lead consumers to purchase both alternatives) whereas prices are set to perfectly segment the consumer market for films with low content durability. We find that lower congestion effects provide studios with incentives to delay release and price the video to induce multiple purchasing behavior. However, an increase in congestion effects can actually lead to higher studio profitability. We also illustrate that, at the lower range of quality, an increase in movie quality should be accompanied by a later video release time. Surprisingly, however, we observe the opposite result at the upper range of movie quality: an increase in quality can justify an earlier release of the video. Finally, we explore the effect of piracy on studio strategy, finding a day-and-date strategy works better for plain vanilla video offerings such as video-on-demand and reduced-feature DVDs while a delayed release strategy can increase profits for feature-rich Blu-ray offerings.

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1 Introduction

Over the past decade, a dramatic change has taken place in how movie studios manage film distribution. In those years, the average time between the initial theatrical release of a film and its debut on digital video has dropped forty percent – from 179 days in 1999 to only 103 days so far in 2013 (Tribbey 2013). This forward shift of 2.5 months has largely been driven by advances in technology, especially video quality and home theater capabilities, but also by an industry that is now challenging antiquated norms (Grover 2005, Cole 2007). Is it profit-maximizing for a release window between sequential distribution channels *always* to exist? In recent years, more films have been skipping theatrical release entirely and going directly to home video as part of a direct-to-video strategy (Barnes 2008); in some cases, studios are even entertaining “day-and-date” strategies which strike at the heart of the matter (Miller 2012, Tartaglione 2013). A day-and-date strategy typically means that a product is released across two or more distinct channels on the same day.¹ For example, in 2006, the film *Bubble* was simultaneously released across all channels by 2929 Entertainment, a company founded by Mark Cuban and Todd Wagner that has vertically integrated across production, distribution, and exhibition – an opportune proving ground for testing such strategies (Kirsner 2007). In fact, many argue that release windows are inherently inefficient since the positive impacts of early promotional spending are not fully captured (Gross 2006). Disney CEO Robert Iger even comments that a film should be released faster on digital video since it has “... more perceived value to the consumer because it’s more fresh” (Marr 2005); perhaps not surprisingly, Disney announced an early video release of *Alice in Wonderland* only 12.5 weeks after its theatrical release instead of the typical 16.5 weeks at the time (Smith and Schuker 2010). Vogel (2007) predicts that further changes in studios’ sequential distribution strategies will continue to occur; as a result, there is a strong need for new research aiming to provide a better understanding of these strategies.

Although the release window is gradually narrowing and a few films have been released using day-and-date strategies, it remains difficult to ascertain the impact of a substantial reduction in the release window on profitability. From an empirical standpoint, the average release window is still approximately four months, and there is very little data on any films with windows ranging from zero to three months. In prior studies, researchers have also commented on the low variance observed in the release window measure (see, e.g., Lehmann and Weinberg 2000). Thus, accurately predicting the effect of releasing a film simultaneously on video or even one month after theatrical release continues to be quite difficult, although some studies have sought to close this gap using surveys (Grover 2006, Hennig-Thurau et al. 2007).

However, one can gain significant insights into how various release strategies would tend to affect

¹In this paper, our focus is on theatrical release and the video market; hence, “day-and-date” is used to signify that the video is released on the same day as the film appears in theaters. We use the term “video” since particular technologies, e.g., Betamax and VHS, have had a tendency to be short-lived. Although “video” currently refers to DVD and Blu-ray products, our model and results will apply regardless of the implementation of these home viewing options, including video on-demand.

Year	Release Window (Days)	Year	Release Window (Days)
1998	200.4	2006	129.2
1999	179.1	2007	126
2000	175.7	2008	127.8
2001	165.4	2009	123.2
2002	171.4	2010	121.4
2003	153	2011	120.2
2004	145.8	2012	114.7
2005	141.8	2013	103.4

Table 1: Shrinking of industry average video-release window from 1998 to 2013 (Tribbey 2013).

consumption and hence profitability if we enhance our understanding of the economic trade-offs consumers face when choosing between theatrical and video alternatives. In this paper, we take a normative approach to studying the theater-video windowing problem. We model the primary economic incentives of consumers making film consumption decisions and subsequently analyze how consumers behave, in equilibrium, as the video release time is varied over its full span. By taking into account strategic consumer behavior, we can explore how studios should set the video release time and price based upon market conditions, movie characteristics, and operational factors. Although consumption by moviegoers can be affected by a wide range of other considerations, in our model we restrict our attention to four primary considerations: *(i)* quality decay of the film content over time, *(ii)* the quality/price gap between theatrical and video alternatives, *(iii)* residual value of the video alternative in addition to movie consumption (multiple purchases), and *(iv)* theater congestion.

For example, all else being equal, consumers prefer to see a film earlier rather than later (Marr 2005). Whether the content is viewed in a theater or at home on video, due to “buzz” generated by marketing, film critics, and social circles, consumers derive highest perceived value at launch, which then decreases over time (Thompson 2006, Cole 2007, Smith and Schuker 2010). Since consumers who watch a film in theaters might also purchase videos of the film, the studios can affect such multiple purchasing behavior by moving up the video release date and pricing it accordingly (Moul and Shugan 2005).

On the other hand, there are two effects that can incentivize viewers to postpone consumption. First, if a film is sequentially distributed through separate channels, some consumers may prefer to wait for lower prices in the secondary channel even though the value derived from the film decays during that time. Thus, a *substitution effect* tends to shift consumption later due to lower prices in the subsequent channel. Second, a *congestion effect* can also shift consumption later in time. Many popular films sell out screenings right after theatrical release. In 2008, *The Dark Knight* was selling out so many screens at midnight on opening day that exhibitors hurried to boost capacity by adding screenings at 3 a.m. and 6 a.m. (Cieply 2008). Furthermore, when Michael Jackson’s comeback

concert footage was assembled into a film titled, *This Is It*, it sold out over one thousand screens (Jurgensen 2009). Anticipating sold-out screenings, some consumers may prefer to delay their viewing. However, a film need not sell out to induce consumers to delay. Even higher utilization of theaters leads to longer waits at ticketing and concessions, poorer seat choices in auditoriums, and other undesirable crowd externalities (e.g., crying babies, cell phones, talking, temperature problems, discomfort, and more). Hence, some consumers may avoid viewing a movie at the box office due to congestion-generated problems. Because congestion also strongly interacts with the studio’s choice of video release time and its price due to substitution effects, optimal management of the entire system requires careful coordination.

In this paper, our research objective is to develop an understanding of how a studio should coordinate the video release time and price for its film as part of a comprehensive strategy to manage the film’s total profitability across the theater and home video channels. Because earlier release times greatly impact cannibalization, it is critical that we capture the preferences of consumers who make multiple purchases, i.e., pay to see the film in a theater as well as purchase the video, as part of the central trade-off. Such preferences tend to mitigate cannibalization losses, but we also parameterize the delay after which multiple purchasing consumers actually obtain residual value to study regions of limited mitigation. Because delaying consumption due to congestion becomes more appealing with earlier video release times, we capture different classes of congestion sensitivity in the consumer population. Using our model, we fully characterize market conditions under which direct-to-video, day-and-date, perfect segmentation, and delayed release strategies maximize profitability for a studio. In each case, we provide an analytical characterization of the optimal video release time and price, and offer studios practical insights on how to utilize each lever to shape consumer demand toward preferable outcomes. Additionally, we suggest how studios can manage piracy concerns by adapting their release strategy, as well as utilize congestion during seasons of peak movie release to actually increase profitability of certain types of films.

2 Literature Review

Eliashberg et al. (2006) provide an extensive review of research related to the motion picture industry. In their discussion of the distribution stage, they pose several questions in regard to the substitutability of DVDs for theatrical consumption and how consumers make trade-offs between the two product forms. We investigate these topics at the consumer level in order to study various video release strategies while generating broader implications on how to optimally manage sequential distribution; thus, our work is closest in nature to research that examines the time window between theatrical and video release.

Frank (1994) studies the timing of sequential distribution by constructing a theoretical model in which potential revenue functions for both product forms linearly decrease in time. Minimizing the sum of the opportunity cost of a video release in the theatrical market and the opportunity

cost of a delay of the video release, he characterizes the optimal, positive video release time. Lehmann and Weinberg (2000) construct a reduced-form model in the context of a video rental firm managing inventory ordering decisions and also analytically characterize the optimal release. Studying potential consumption of both versions in a model of intertemporal movie distribution, Calzada and Valletti (2012) show that versioning can be optimal for information goods with zero marginal costs. They further establish that a monopolist, or a centralized channel, will often price and simultaneously release both versions whereas a sequential release strategy may be driven by the vertical, decentralized channel structure found in the motion picture industry. Our paper complements this stream of literature by explicitly considering the following three factors on the studio’s optimal video release time and pricing strategy: *(i)* an endogenously arising congestion externality at theaters with heterogeneity in consumer sensitivity to congestion; *(ii)* a time delay after which consumers who buy both versions obtain residual value associated with the second purchase; and *(iii)* a decaying quality over time.

The empirical analysis in Luan and Sudhir (2006) is also related to our work since they account for buzz decay and multiple purchases in their utility specification. In their study, they find that both highly rated films and animated films tend to be less substitutable and that, on average, the optimal release window should be 2.5 months. In another empirical study, Hennig-Thurau et al. (2007) examine multiple channels and release orderings. Although the introduction of DVD rentals is country-dependent, they similarly find that DVD sales should optimally be delayed by three months. Nelson et al. (2007) study the time gap between the end of a film’s theatrical run and its release on DVD, finding that about 30 percent of films have DVD versions released while the film is still in theaters and that the time gap is generally declining.

Our work is also related to papers that study the impact of a social presence on consumers. Harrell et al. (1980) find that perceived crowding negatively affects shopping behavior as consumers employ adaptation strategies. Hui et al. (2009) study shoppers’ path behavior and zone density, finding that consumers might be attracted to higher density zones but shop less in them. Argo et al. (2005) demonstrate that increasing social presence tends to positively affect emotions initially and then to have a more negative effect as the presence gets larger. In line with their finding, we focus on the impact of congestion on the theater-viewing experience since theater owners maximize their profits by allocating screens based on movie demand. Specifically, theater owners have economic incentives to keep their capacity highly utilized, and congestion-sensitive consumers are likely to be negatively affected at these higher levels of congestion.

In this paper, we build on the ideas summarized above with the goal of developing a theoretical framework in which we start from the consumer’s choice problem in order to clarify the trade-offs consumers make and how demand for each product form arises as the result of equilibrium strategic behavior. Our research goals are normative in nature as we seek to explore how consumers should behave as the release time is manipulated across its full span, and, with this understanding, how studios should optimally adjust the timing and pricing of video releases. In particular, our work is

complementary to the above papers because, using our model, we can further examine the impact of smaller release windows and even the viability of day-and-date strategies. Because, historically, these windows have each lasted several months, most models have not explored the simultaneous release of a film in both theatrical and video forms (see, e.g., Lehmann and Weinberg 2000). Furthermore, the lack of sufficient data makes it difficult to estimate the impact of day-and-date strategies.²

The observable facts are that release windows have continued to shrink over the last decade (Cole 2007, Tribbey 2013) and that day-and-date strategies are currently being tested (Abramowitz 2004, Hofmann 2005). Thus, there is now a strong need to develop a sound theoretical understanding of the effects of such strategies. By utilizing an approach that captures individual consumer decision-making while accounting for primary influential factors (including congestion costs, quality decay, product substitutability, and content durability), we provide insights into how release timing and pricing affects which product forms are consumed and the net effect on profitability. Further, our paper provides a deeper understanding of sequential distribution management and a foundation for further studies that can explore how to shape demand by targeting consumer incentives.

In a broader context, sequential product introduction is related to intertemporal price discrimination (see, e.g., Coase 1972, Bulow 1982, Gul et al. 1986, Besanko and Winston 1990, and Desai et al. 2004). The papers in this literature demonstrate that a firm competes against itself by selling across different time periods without commitment. Exploring intertemporal pricing under capacity constraint, Su (2007) shows that markdown pricing can be optimal when high valuation customers are less patient and that, otherwise, optimal prices are increasing in time.

Moorthy and Png (1992) consider a seller with two substitutable and differentiated products and show that sequential introduction can both effectively reduce cannibalization and be more profitable than simultaneous introduction. Studying films with short runs, Waterman and Weiss (2010) find that the release window for videos is still long and invariant to theatrical run length. Their results suggest that studios can credibly commit to release windows. We examine sequential introduction where, upon theatrical movie release, the studio commits to a video release time and price.

3 Model

There is a continuum of consumers who are heterogeneous in their sensitivity to the quality of a cinematic production. Each consumer's sensitivity (i.e., her type) is uniformly distributed on $\mathcal{V} \triangleq [0, 1]$. We assume that the product can be consumed in theaters (the *movie*) for a given price $p_m > 0$ at time zero, and consumed in digital form at home (the *video*) for $p_d > 0$. In the

²As of March 2013, there are approximately 17 films that have been released through a secondary distribution channel within two weeks of theatrical release worldwide. Ten of these films have been released in theaters and through a secondary distribution channel on the same day. In addition, distributors in Europe will begin testing the viability of day-and-date releases as a part of the TIDE Experiment during the summer and fall of 2013 (Goodfellow 2013).

movie industry, there are many other channels for obtaining film content including pay-per-view on-demand services (e.g., Vudu), video rental services (e.g., Netflix, Blockbuster, and Redbox), and cable services (e.g., Time Warner and Comcast). Our model simplifies this setting and focuses on clarifying the main trade-offs between a primary and secondary channel. However, the insights derived from our analysis can be readily applied to the more complex setting.

Two of the primary factors identified in Section 1 that affect consumption are the quality/price gap between alternatives and quality decay. First, to capture the former factor, we adapt a standard model of vertical product differentiation to our specific setting (Shaked and Sutton 1983, 1987). The product consumed in theaters has an inherent level of quality given by $\gamma_m > 0$. For example, the box office hit *Avatar* would be associated with a higher value for γ_m due to its special effects and 3-D features. If a consumer with quality sensitivity $v \in \mathcal{V}$ views the movie in a theater, her maximum willingness to pay is given by $\gamma_m v$. Similarly, the inherent quality of the corresponding video is given by $\gamma_d > 0$. Second, as discussed earlier, the content of the film loses its value over time. In that sense, we can consider the video to be essentially a different product at each moment in time.³ If the video is released at time $T \in [0, 1/\beta]$ and consumer v prefers to only purchase the video, her willingness to pay is $\gamma_d(1 - \beta T)v$, where β is the film's quality decay rate.⁴

The third critical factor we identified relates to the residual value of the video for consumers who make multiple purchases. Because consumers often purchase videos of films they have already seen in theaters, it is important to permit consumption of both movie and video alternatives in the model. Should a consumer opt for multiple purchases by consuming both the movie and the video, her willingness to pay for the video is modified to $\delta\gamma_d(1 - \beta \max(T, \xi))v$, where $\delta \in [0, 1]$ represents the *durability* of the film in terms of its content, and $\xi \in [0, 1/(2\beta)]$ denotes the minimum time beyond which the consumer is again willing to pay for the durable content. For example, a larger value for δ indicates that consumers still derive considerable value from watching the film on video again after viewing it in a theater. At the boundary, i.e., when $\delta = 1$, a consumer deciding whether to purchase the movie or video will consider each form of the product independently.⁵ However, when $\delta < 1$, the video serves as a substitute for the movie since the residual value associated with multiple purchasing diminishes. That is, smaller values for δ imply that a consumer has less incentive to consume both alternatives; instead, she tends toward the one providing higher net utility. In particular, at $\delta = 0$, a consumer purchases at most one form of the product: either a movie in a theater or a video, but not both.

³Note that the focus is not on how consumers discount the timing of consumption for the *same* product (e.g., exponential or hyperbolic discounting considerations). In our case, there is substantial decay in the quality of the film itself over time not because the content has changed but because it is steadily losing its relevance. Thus, in our model we focus on this aspect of quality decay over time and disregard discounting concerns that are relatively insignificant for film windows and, even if included, would not affect our main insights.

⁴We employ a linear quality decay in time for both alternatives. Our results do not critically depend on this functional form, and we model quality decay in this manner primarily for simplicity and tractability. For example, Luan and Sudhir (2006) similarly utilize this form for quality decay in their utility specification.

⁵Specifically, when $\delta = 1$, a consumer still derives full value from viewing the content on video even though she also saw the same content in a theater. Because there is no loss in value, she can treat each decision independently. We restrict attention to $\delta \in [0, 1]$ since $\delta > 1$ is unlikely, and the intuition gained by studying high δ will still apply.

In our model, the content durability is a film characteristic, but it directly interacts with consumers’ heterogeneous types. Specifically, consumers with the highest types are the ones with the greatest incentive to engage in multiple purchases. However, the durability of the content itself is dependent on the type of film produced. For example, children’s movies such as *Pocahontas*, *Aladdin*, and *Cars*, would likely be associated with a higher δ because their content maintains large residual value for repeated viewings. Similarly, films that have established subcultures (e.g., *Star Wars* and *Star Trek*) would also have higher durability. On the other hand, documentary films and historical dramas like *Hotel Rwanda*, in which the focus lies on being informative, may have relatively lower residual value after a first viewing in comparison to highly entertaining films. Also, emotional dramas and films with high cognitive costs such as *The Machinist*, *Leaving Las Vegas*, and *The Blair Witch Project* tend to exhibit lower content durability (Kinnear 2012).

We can now specify the timing of decisions and formally define the consumer strategy set. At the beginning stage, the production studio determines when to open its video distribution channel and sets the video price. The studio announces and commits to this video release time which is denoted by T as well as the video price p_d . Subsequent to the announcement, each consumer decides whether to consume only the movie ($s = M$), only the video ($s = D$), both the movie and video ($s = B$) or neither alternative ($s = N$). The strategy set is thus denoted by $S \triangleq \{M, D, B, N\}$, and each consumer chooses the action $s \in S$ that maximizes her payoff.

When the product is consumed in a theater (i.e., either $s = M$ or $s = B$), congestion externalities arise due to the theater’s fixed capacity and limited resources. For example, as the number of patrons seeing a movie at a theater grows large, there is an increased risk of screenings being sold out, having only poor seats remaining, and the viewing experience being degraded due to congestion externalities. Congestion is the fourth factor we capture in our model that critically affects moviegoer consumption.⁶ We use the term “congestion” in a vein similar to that of Vickrey (1955) in his study of New York City’s subway system where he states, “...where congestion occurs, the fare may fail to reflect the relatively high cost either of providing additional service at such times, or of the added discomfort to existing passengers occasioned by the crowding in of additional passengers.” We highlight this point since traditional congestion costs in the operations management literature stem from longer waiting times in service processes. In our context, however, consumers do not simply wait at theaters and incur costs until a screen becomes free; rather, they adapt by altering their consumption decision. For these reasons, we model congestion costs as proportional to the mass of consumers viewing the movie in a theater. Ceteris paribus, a consumer strives to consume earlier to increase her surplus due to quality decay, but congestion provides incentives to delay and substitute to alternative content forms.

We let $\sigma : \mathcal{V} \times \mathcal{C} \rightarrow S$ be a strategy profile of consumer actions and denote the mass of consumers

⁶We conducted a survey and found that 45% out of 209 respondents of the moviegoer population chose either “Very likely to delay” or “Definitely will delay” in response to a question of whether they would consider delaying film consumption because of crowds at theaters.

choosing in-theater consumption with $\eta(\sigma)$.⁷ Because consumers may vary in their sensitivity to congestion, we examine two classes of consumers: $\mathcal{C} = \{H, L\}$. Class H refers to consumers who are sensitive to congestion, and thus may have a positive congestion cost parameter denoted as $\alpha \geq 0$. A class H consumer with quality sensitivity v obtains a net payoff of $\gamma_m v - \alpha \eta(\sigma) - p_m$ if she consumes the movie only (i.e., $\sigma(v, H) = M$), for example. On the other hand, a class L consumer is less sensitive to congestion and for convenience we assume her congestion parameter is zero such that her net payoff analogously becomes $\gamma_m v - p_m$. Finally, we denote the probability any given consumer $v \in \mathcal{V}$ belongs to class L and H with $\rho \in [0, 1]$ and $1 - \rho$, respectively. Fixing all other consumers to the strategy prescribed by σ_{-v} , we can summarize the net payoff to the consumer with quality sensitivity v when undertaking action s by:

$$V(v, H, s, \sigma_{-v}) \triangleq \begin{cases} \gamma_m v - \alpha \eta(\sigma) - p_m + \delta \gamma_d (1 - \beta \max(T, \xi)) v - p_d & \text{if } s = B; \\ \gamma_m v - \alpha \eta(\sigma) - p_m & \text{if } s = M; \\ \gamma_d (1 - \beta T) v - p_d & \text{if } s = D; \\ 0 & \text{if } s = N, \end{cases} \quad (1)$$

for class H , and

$$V(v, L, s, \sigma_{-v}) \triangleq \begin{cases} \gamma_m v - p_m + \delta \gamma_d (1 - \beta \max(T, \xi)) v - p_d & \text{if } s = B; \\ \gamma_m v - p_m & \text{if } s = M; \\ \gamma_d (1 - \beta T) v - p_d & \text{if } s = D; \\ 0 & \text{if } s = N, \end{cases} \quad (2)$$

for class L . We focus our study on the more commonly-observed parameter region where $\gamma_m > \gamma_b$ satisfied so that the inherent quality of the theatrical experience is higher than the video. For example, going to the theater can be thought of as a complementary event that includes viewing the film and thus carries a higher quality. We also focus on the region where $\gamma_m > p_m$ is satisfied such that there exist consumers who can obtain positive surplus at the theater. Finally, we assume that the movie price p_m is high enough that the video pricing decision is not constrained above. Later, we relax this assumption and examine the impact of a video pricing constraint which can potentially arise as studios look to deter piracy.

4 Consumer Market Equilibrium and the Studio's Problem

In this section, first taking the video release time T , video price p_d and other model parameters as given, we derive the consumer market equilibrium. Its characterization consists of two components. We can classify consumers by the product forms they consume: both the movie and the video

⁷Consumers form rational expectations on $\eta(\sigma)$ which are fulfilled in equilibrium. In equilibrium, $\eta(\sigma)$ is equal to the theatrical movie demand.

(*both*), only the movie (*movie*), only the video (*video*), or nothing (*none*).⁸ First, we develop an understanding of what types of consumption outcomes occur under the various market conditions. For example, if a blockbuster movie is coupled with a fast video release time, to what extent will theater demand be cannibalized, particularly for high content durability films? By gaining insight into how moviegoers adjust their consumption patterns in response to durability, video release times, and congestion, we can more clearly see how release timing and pricing affect profitability, a subject we address in the next section.

Thus, taking into account a film's quality decay over time, a congestion externality, and the availability of a video alternative, each consumer chooses an action that maximizes her own surplus. An equilibrium strategy profile σ^* must satisfy the following for each $v \in \mathcal{V}$ and $c \in \mathcal{C}$:

$$V(v, c, \sigma^*(v, c), \sigma_{-v}^*) \geq V(v, c, s, \sigma_{-v}^*) \quad \text{for all } s \in S. \quad (3)$$

We begin by characterizing the general, threshold structure of the equilibrium strategies.

Proposition 1 *There exist threshold values ω_b^c , ω_m^c , $\omega_d^c > 0$ (where $c \in \mathcal{C}$ refers to the consumer class) such that the equilibrium consumer strategy profile is given by*

$$\sigma^*(v, c) = \begin{cases} B & \text{if } \omega_b^c \leq v \leq 1; \\ M & \text{if } \omega_m^c \leq v < \omega_b^c; \\ D & \text{if } \omega_d^c \leq v < \omega_m^c; \\ N & \text{if } v < \omega_d^c. \end{cases} \quad (4)$$

Proposition 1 formally establishes that the equilibrium consumer strategies are ordered such that (i) consumers with the lowest sensitivity to quality (i.e., low types) remain out of the market; (ii) consumers with slightly higher quality sensitivity purchase only the video alternative; (iii) consumers with even higher sensitivity choose to view the movie in theaters; and finally (iv) consumers with the highest quality sensitivity consume both the movie and video alternatives. This threshold structure holds for both consumer classes, H and L , who vary with regard to congestion costs.

Whether any particular consumer segment: *both*, *movie*, *video*, or *none* is present in equilibrium critically hinges on the underlying parameter region. For example, as we previously discussed, congestion is a significant factor in film consumption choice, and controlling congestion is frequently an important objective, particularly in traditional service processes. In our setting, congestion is the result of aggregate consumers' purchasing decisions. However, unlike traditional service processes, the costs are not due to waiting to be serviced; here, consumers can freely consume the movie now or purchase the video instead. Rather, the costs are associated with discomfort due to highly utilized theaters (Vickrey 1955). Thus, each individual consumer must strategically select her own optimal consumption strategy while taking others' strategic behavior into account. When the congestion

⁸Throughout the rest of the paper, we use the italicized words: *both*, *movie*, *video*, and *none* to refer to each of these specific consumer market segments.

cost parameter is high for class H consumers, we will establish that $\omega_b^H = \omega_m^H = 1$ such that, by (4), only *video* and *none* segments arise in $\sigma^*(v, H)$ in equilibrium.

Taking into consideration the equilibrium consumer strategies developed above, we next lay out the decision problem faced by studios. We denote the demand for the movie by

$$D_m \triangleq \int_{\mathcal{V}} [\rho \mathbf{1}_{\{\sigma^*(v, L) \in \{B, M\}\}} + (1 - \rho) \mathbf{1}_{\{\sigma^*(v, H) \in \{B, M\}\}}] dv, \quad (5)$$

which measures the population of consumers whose equilibrium strategy includes viewing the film in a theater. Similarly, we denote the aggregate demand for the video by

$$D_d \triangleq \int_{\mathcal{V}} [\rho \mathbf{1}_{\{\sigma^*(v, L) \in \{B, D\}\}} + (1 - \rho) \mathbf{1}_{\{\sigma^*(v, H) \in \{B, D\}\}}] dv. \quad (6)$$

We denote the studio's share of movie and video revenues with λ_m and λ_d , respectively, where $0 \leq \lambda_m, \lambda_d \leq 1$. Since the marginal cost of satisfying consumers, whether in a theater or by providing a video, is fairly small, we make a simplifying assumption that it is zero. Thus, the studio's profit function can be written as

$$\Pi(T, p_d) \triangleq \lambda_m p_m D_m + \lambda_d p_d D_d. \quad (7)$$

The main objective of this paper is to develop an understanding of how consumers adapt their consumption choices to changing video release times and price, and, subsequently, to characterize how studios can manage them by optimizing video release time and pricing. Hence, we take movie prices as fixed and exogenous to the model.⁹ The studio's problem can then be written as

$$\max_{T \in [0, 1/\beta], p_d > 0} \Pi(T, p_d) \quad (8)$$

$$s.t. \quad \sigma^*(\cdot | T, p_d) \text{ satisfies (3).}$$

As can be seen in (7), prices have a direct effect on studio profits while all parameters, including release time and prices, indirectly influence profitability through their impact on the strategic consumption behavior of consumers. Finally, social welfare is defined

$$W = p_m D_m + p_d D_d + \int_{v \in \mathcal{V}} [\rho V(v, L, \sigma^*(v, L)) + (1 - \rho) V(v, H, \sigma^*(v, H))] dv. \quad (9)$$

To provide further insight into the release timing problem faced by the studio, we next illustrate how both studio revenues and the consumer market equilibrium are affected by changes in T given p_d . In Figure 1, which illustrates how release time affects consumer market structure and profitability, we examine the case where film durability falls within an intermediate range (e.g., $\delta = 0.4$ in the plot). In panel (c), we plot the typical shape of the studio's profit curve. Panels

⁹Each year, the Motion Picture Association of America compiles statistics on the theatrical market. In the last 5 years, the average domestic movie ticket price has risen from \$7.18 in 2008 to \$7.96 in 2012, which is fairly stable when accounting for changes in the consumer price index over the same time range (MPAA 2011).

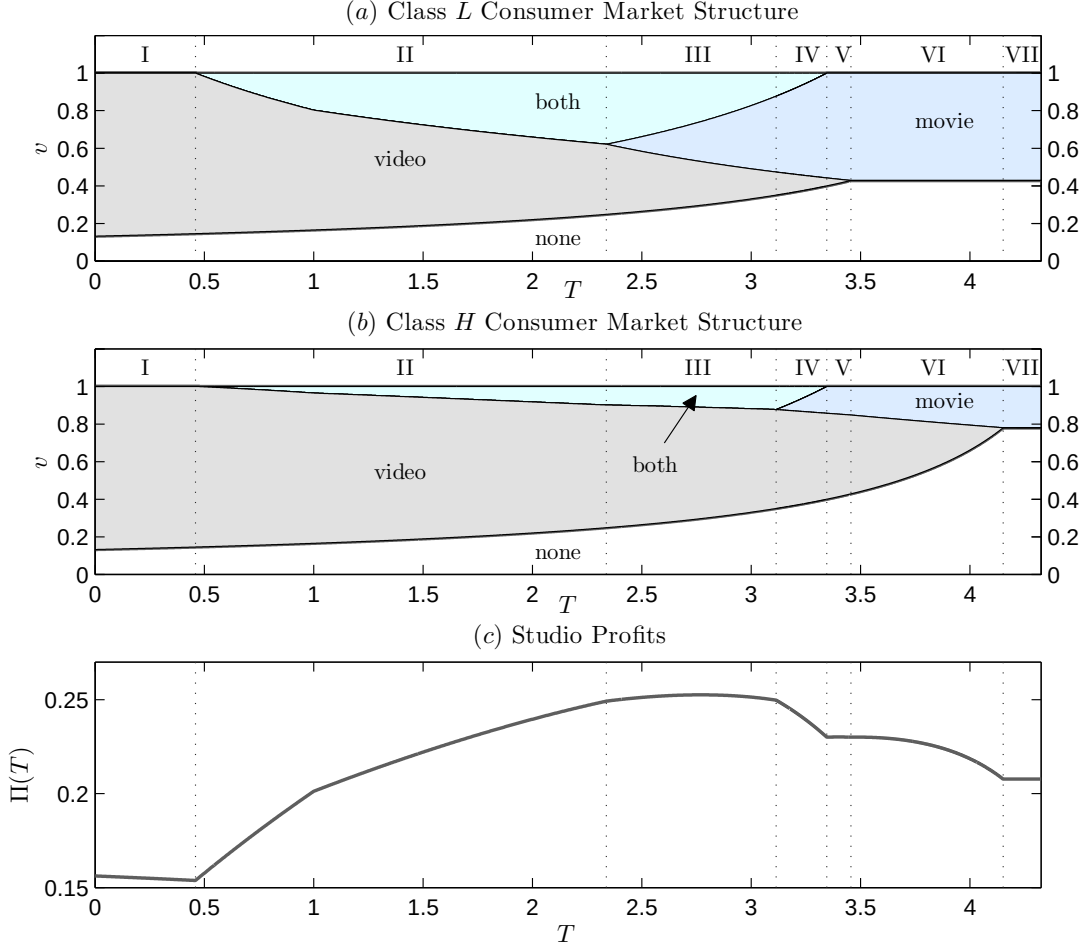


Figure 1: How the video release time affects the studio's revenue and the consumer market structure. In panel (a), class L consumers' optimal strategies are identified. In panel (b), class H consumers' optimal strategies are plotted. In panel (c), the studio's profit is plotted as a function of video release time. The parameter values are $\alpha = 4$, $\beta = 0.2$, $\delta = 0.4$, $\gamma_m = 3.5$, $\gamma_d = 3.4$, $\lambda_m = 0.45$, $\lambda_d = 0.4$, $p_m = 1.5$, $p_d = 0.45$, $\rho = 0.25$, and $\xi = 1$.¹¹

(a) and (b) show the corresponding equilibrium consumer market structures for classes L and H , respectively, also as a function of the release time to demonstrate their underlying effect on revenue. When T is low (Region I) and the video is a close substitute for the movie in terms of quality, consumers are not willing to make multiple purchases provided that δ is not too high. Consequently, the resulting consumer market structure, $\sigma^*(v, c) \in \{D, N\}$ for all $c \in \mathcal{C}$, is characterized by a sizable population of consumers who only purchase the video, as depicted in Region I of panels (a) and (b). Within this region, an increase in T only reduces the value of the video offering, which in turn causes profits to decrease, as is indicated in the corresponding portion of panel (c).

As the video release time further increases to Region II in panels (a) and (b), the quality

¹¹In all figures presented in this paper, the set of numerical parameters listed in the caption reflects the conditions of the proposition being illustrated. As formalized in the proposition statements, the essence of each illustration will hold over wide parameter ranges and in no way critically depends on the representative set of parameters employed.

of the video is decayed so that the most quality-sensitive consumers now prefer to watch the movie in theaters as well as on the video, which then results in the consumer market structure $\sigma^*(v, c) \in \{B, D, N\}$ for all $c \in \mathcal{C}$. This shift in the consumer market structure is associated with increasing profits in Region II due to the increase in the *both* segment as is shown in panel (c). As T further increases into Region III, a *movie* segment from class L consumers begins to arise. In Region IV, a *movie* segment from class H consumers also arises in equilibrium. In these regions, as the consumer market begins to sharply lose the *both* segment and *video* segment with a later release time, revenues diminish rapidly starting at $T > 3.1$. In Regions V and VI, which initially correspond to the consumer market structure $\sigma^*(v, c) \in \{M, D, N\}$ for both classes of consumers, a further increase in T can again have a profitable effect because it can now induce video purchasers to become movie viewers; the profitability of this substitution effect is illustrated in the corresponding region of panel (c). Finally, for sufficiently large T (in Region VII), the video is no longer a rational alternative and the studio's profits are no longer affected by changes in release time.

Note that when content durability (δ) is fairly low, it is relatively difficult for the studio to induce multiple purchases; in this case, the market structures corresponding to Regions II, III and IV from panels (a) and (b) would not occur. As a consequence, the studio can either release the product at time zero (Region I) or choose a release time sufficiently far out to operate in an analog of Regions V, VI and VII. This difference highlights how film durability affects the studio's release time decision.

5 Optimal Release Time and Pricing

In Section 4, we described equilibrium consumption behavior, formulated the studio's optimal video release time and pricing problem, and then demonstrated how release timing interacts with the consumer market structure and profitability. Having built up this understanding, we next examine how studios select among direct-to-video, day-and-date, delayed release, and other channel strategies to maximize returns associated with their productions. We characterize the market environments under which these diverse strategies are effective as well as how an appropriate video release time and pricing strategy can be tailored to fit film-specific marketing and operational characteristics.

We begin with the direct-to-video strategy. With this strategy, studios set video release time and price such that the equilibrium consumer market structure satisfies

$$\sigma^*(v, c) = \begin{cases} D & \text{if } \frac{p_d}{\gamma_d(1-\beta T)} \leq v < 1; \\ N & \text{if } v < \frac{p_d}{\gamma_d(1-\beta T)}, \end{cases} \quad (10)$$

for all $c \in \mathcal{C}$, which is to say that only the *video* and *none* consumer segments are represented. It is worth noting that if the studio ever chooses T such that an equilibrium characterized by (10) arises, the studio would be strictly better off by setting $T=0$ since the consumer market size is

completely governed by the lower bound on quality sensitivity $p_d/(\gamma_d(1 - \beta T))$. By delaying the release time, the studio only loses video demand with no gain. In the following proposition, we present the conditions under which this equilibrium structure is optimally induced by studios.

Proposition 2 (Direct-to-Video Strategy) *When the quality-to-price ratio of the theatrical offering is sufficiently low, a direct-to-video strategy is optimal ($T^* = 0$) and the video price satisfies $p_d^* = \frac{\gamma_d}{2}$.*

When the quality-to-price ratio for the movie is low, it becomes difficult for a studio to create moviegoing incentives without either severely reducing the quality of its video alternative by substantially delaying release or increasing the video price quite high. Proposition 2 highlights the fact that the substitution effect can be sufficiently strong that no consumer finds it optimal to consume the movie in a theater, even in the absence of any congestion effects, over a broad range of video release times and prices. In such a case, a studio will find it profitable to maximize the quality of its video offering by releasing it immediately and setting a price both consistent with the video's quality and reflecting the studio's monopoly power over its own content. For such an equilibrium outcome to arise, the video should be strongly preferred over the theatrical movie, which occurs when a lower-quality, niche film is released early.

In the film industry, the number of direct-to-DVD films has grown 36 percent since 2005 with 675 films being released in 2008, according to Adams Media Research, and the direct-to-DVD market generates approximately \$1 billion in annual revenues (Barnes 2008). Oftentimes, studios pursue a direct-to-video strategy for films with lower quality-to-price ratios. Further, we observe both high and low durability films taking this path. As an example of highly durable content, Disney often uses a direct-to-video strategy for lower quality, follow-up sequels to its blockbuster animated films. The sequel to *Aladdin* (1992) called *Return of Jafar* (1994) had a production budget of only \$5 million in comparison to *Aladdin's* \$28 million budget and, using a direct-to-video strategy, it generated wholesale revenues of \$120 million (Vogel 2007); Disney found similar success with *The Lion King 1 1/2*. Movie studios also employ a direct-to-video strategy with lower quality sequels of films with likely less content durability. For example, Universal Studios released follow-up low budget films: *American Pie Presents: Band Camp*, *American Pie Presents: Beta House*, and *American Pie Presents: The Book of Love*, capitalizing on a niche market of faithful consumers of their *American Pie* series.

We next consider the case in which the quality-to-price ratio of the theatrical offering is high. In this case, the theatrical movie offering becomes a lucrative channel for the studio. We first study the region with a high congestion cost parameter. Because of theaters' capacity constraints, the congestion parameter can often be high, particularly during periods when big-budget movies are released (e.g., Memorial Day, Fourth of July, Thanksgiving, and Christmas) as these films compete for a limited number of screens (Einav 2007). In this region, if content durability δ is high, consumers who belong to the less congestion-sensitive class ($c = L$) now have greater incentives

to consume either *both* or the *movie* offerings. In particular, the consumer equilibrium structure in this region for class L tends toward

$$\sigma^*(v, L) = \begin{cases} B & \text{if } \frac{p_d}{\delta\gamma_d(1-\beta\max(\xi, T))^+} \leq v < 1; \\ M & \text{if } \frac{p_m}{\gamma_m} \leq v < \frac{p_d}{\delta\gamma_d(1-\beta\max(\xi, T))^+}; \\ N & \text{if } v < \frac{p_m}{\gamma_m}. \end{cases} \quad (11)$$

Moreover, consumers who belong to the more congestion-sensitive class ($c = H$) are incentivized to forgo consumption in theaters and instead purchase either the video option or nothing, i.e., $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$, as given in (10), due to the size of the equilibrium congestion externality. In this scenario, the optimal release strategy for the studio is a day-and-date strategy which we formalize in the following proposition:

Proposition 3 (Day-and-Date Strategy) *If the quality-to-price ratio of the theatrical offering, content durability and congestion cost parameter are sufficiently high, then a day-and-date strategy is optimal ($T^* = 0$) and the video price satisfies $p_d^* = \frac{\delta\gamma_d(1-\beta\xi)}{2(\delta(1-\beta\xi)(1-\rho)+\rho)}$.*

When the content durability δ of a film becomes sufficiently high, the movie demand D_m in equilibrium becomes $\rho(1 - p_m/\gamma_m)$, which is independent of the video release time T . Moreover, the video demand D_d monotonically decreases in the release time. Consequently, an immediate video release ($T^* = 0$) is optimal for the studio for any video price. Notably, in this case, a strategic video release at $T^* = 0$ still gives rise to in-theater demand ($D_m > 0$) from class L consumers. Because movie demand is independent of p_d and video demand does not depend on γ_m and p_m (i.e., the associated revenues from each are separable), the optimal video price p_d^* is solely determined by maximizing video revenue.

As can be seen in the characterization of p_d^* in the statement of Proposition 3, when video quality increases, either directly via γ_d or from residual quality through $\delta\gamma_d$ in the *both* segment, the optimal video price increases. In contrast, as the residual quality in the *both* segment decreases due to increases in β or ξ , the optimal video price decreases. When ρ increases, the size of the *both* segment increases in turn. However, a higher ρ reduces the size of the *video* segment from class H consumers, which is the dominant effect. Consequently, as ρ increases, p_d^* decreases.

Proposition 3 implies that a day-and-date strategy is an attractive video release strategy for the studio, particularly for films with high theatrical quality and high content durability, such as blockbuster children's movies. Commenting on the motion picture industry, Walt Disney Co. CEO Robert Iger said, "I don't think it's out of the question that a DVD can be released in effect in the same window as a theatrical release," suggesting that day-and-date video release strategies are being considered. Noting that children's movies often have high content durability and low ξ values, Iger also suggested selling DVDs of *Chicken Little* (2005) in theaters in which the movie was playing in an interview with The Wall Street Journal (Donaldson-Evans 2006).

One issue with employing a day-and-date strategy is potential push-back from theater owners,

who are concerned that early video availability will cannibalize theatrical movie demand. One solution is to sell early released DVDs only in theaters at patron exit areas such that these DVDs can be targeted to the *both* consumer market segment, which will not cannibalize movie demand. Another option that studios may consider to alleviate theater owners' concerns is to implement revenue sharing for video sales with theater owners, which has also been suggested by 2929 Entertainment (Grover 2006). In addition, one ancillary benefit of using a day-and-date strategy relates to advertisement costs, which amount to half of the total production cost on average (Vogel 2007). When a studio releases a film's video a few months after being released in theaters, it must once again incur additional advertising expenditures. Under a day-and-date strategy, a studio can consolidate a film's marketing budget into a single and shorter period, leveraging the initial buzz effectively across both movie and video offerings.

Next, we examine how a film with lower content durability affects the consumer market equilibrium and the studio's release time and pricing problem. Lower content durability reduces the residual value of the video after theatrical consumption, and hence shrinks the size of *both* segment. If δ is sufficiently low, the *both* consumer segment disappears entirely. In this case, the studio achieves perfect segmentation, i.e., all *movie* demand is derived from class L consumers and all video demand stems from class H consumers. In particular, the equilibrium consumer market structure for class L consumers becomes

$$\sigma^*(v, L) = \begin{cases} M & \text{if } \frac{p_m}{\gamma_m} \leq v < 1; \\ N & \text{if } v < \frac{p_m}{\gamma_m}, \end{cases} \quad (12)$$

and the equilibrium consumer market structure for class H is given in (10).

Proposition 4 (Perfect Segmentation) *When the quality-to-price ratio of the theatrical offering and congestion cost parameter are sufficiently high, whereas content durability and the proportion ρ of class L consumers are both low, a studio should release the video immediately ($T^* = 0$) with prices at $p_d^* = \frac{\gamma_d}{2}$ to achieve perfect segmentation.*

Perfect segmentation is a special case of a day-and-date strategy where the video price is set to preclude the existence of a *both* segment in the consumer market equilibrium. When class L consumers forgo the *both* option due to low content durability and a high video price, there does not exist any video demand stemming from class L , as can be seen in (12). Furthermore, due to high congestion costs stemming from the externality caused by class L consumers, there is no movie demand generated by the more congestion-sensitive class H consumers. As a result, the classes separate; class L consumers compose the *movie* segment, and class H consumers compose the *video* segment. With perfect segmentation of consumers, the studio is not concerned with cannibalization of movie demand due to early video release. Therefore, the studio releases the video immediately ($T^* = 0$) using a day-and-date strategy, and sets the corresponding monopoly price for the video ($p_d^* = \frac{\gamma_d}{2}$).

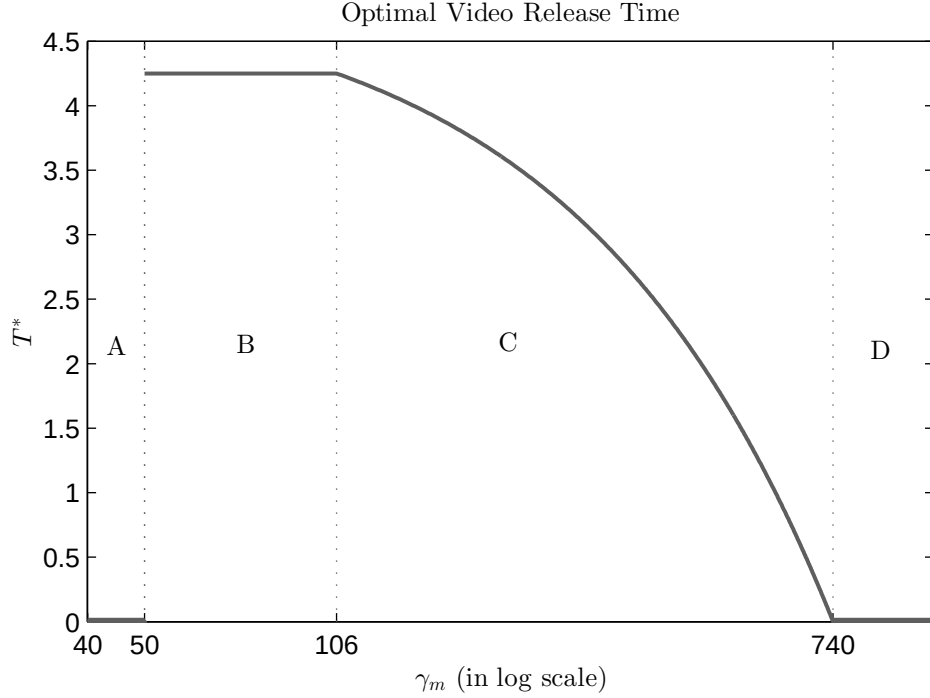


Figure 2: The impact of theatrical quality on the optimal video release time. The parameter values are $\alpha = 10$, $\beta = 2$, $\delta = 0.8$, $\gamma_d = 30$, $p_m = 40$, $\rho = 0.7$, and $\xi = 4.25$.

As we discussed above, *Bubble* was released simultaneously across different channels including a theatrical release, i.e., a day-and-date release strategy was employed. Considering its production budget of \$1.6 million as well as its audience rating, it is a relatively low quality (γ_m) film. Our model suggests using a direct-to-video strategy as the optimal video release strategy for *Bubble*, as indicated in Proposition 2, rather than a perfect segmentation strategy (Proposition 4). Given that its domestic total box-office revenue was less than \$150K,¹² a direct-to-video strategy could have been a more efficient use of theatrical resources, given the low screen utilizations likely obtained during the run length of this movie. Rather, a higher quality movie with higher content durability is a better candidate for experimenting with day-and-date release strategies according to our model.

Figure 2 illustrates the optimal video release time as theatrical movie quality changes for a high congestion cost parameter, summarizing the results in Propositions 2 and 3. If γ_m is sufficiently high, the studio releases the video immediately, utilizing a day-and-date strategy as demonstrated in Proposition 3; this behavior is also illustrated in Region D in Figure 2. In contrast, if γ_m is sufficiently low, the studio also releases the video immediately, however this time as part of a direct-to-video strategy as is established in Proposition 2. This strategy is illustrated in the left-hand portion of Figure 2, labeled as Region A. As γ_m takes intermediate values, as in Regions B and C, the studio delays the video release.

¹²According to boxofficemojo.com, its opening weekend box-office revenue was \$70,664 whereas the overall revenue from all channels for the first week was estimated to be \$5 million, a majority of which was derived from DVD shipments (Gray 2006).

Proposition 5 (Delayed Release Strategy) *When the quality-to-price ratio of the theatrical offering is within an intermediate range, content durability and the congestion cost parameter are sufficiently high, and $\lambda_d < \lambda_m$ and $\gamma_d > \underline{\gamma}_d$ are satisfied, a delayed release strategy is optimal. Specifically, if $\gamma_m/p_m < 2\sqrt{\lambda_m\rho/\lambda_d}$, then $T^* = \xi$ and $p_d^* = \gamma_d(1 - \beta\xi)/2$. Otherwise, $T^* = \frac{1-\delta(1-\beta\xi)}{\beta} - \frac{\gamma_m}{\beta\gamma_d} + \frac{2p_m}{\beta\gamma_d}\sqrt{\frac{\lambda_m\rho}{\gamma_d}}$ and $p_d^* = \frac{\gamma_m + \delta\gamma_d(1-\beta\xi)}{2} - p_m\sqrt{\frac{\lambda_m\rho}{\lambda_d}}$.*

In this region, the theatrical movie quality is high enough so that direct-to-video is not an attractive strategy for the studio, i.e., there exists movie demand that would be foregone. However, the quality is also not high enough for a studio to take a day-and-date strategy. In equilibrium, movie demand is derived solely from the *both* segment in class L , which is to say the consumer market structure of class L is given by

$$\sigma^*(v, L) = \begin{cases} B & \text{if } \frac{p_m}{\gamma_m - \gamma_d((1-\beta T) - \delta(1-\beta \max(T, \xi))^+)} \leq v < 1; \\ D & \text{if } \frac{p_d}{\gamma_d(1-\beta T)} \leq v < \frac{p_m}{\gamma_m - \gamma_d((1-\beta T) - \delta(1-\beta \max(T, \xi))^+)}; \\ N & \text{if } v < \frac{p_d}{\gamma_d(1-\beta T)}. \end{cases} \quad (13)$$

When γ_m increases to Region B in Figure 2, γ_m is not high enough to induce the *both* segment with the video release time being anything smaller than ξ . As a result, in this region, the studio releases the video at $T^* = \xi$ and sets the monopoly video price corresponding to that release time. If γ_m increases further into Region C, the studio can now induce a *both* segment using a release time less than ξ . The studio then sets a video price optimally depending on movie characteristics such as γ_m and p_m through careful coordination with the optimal video release time T .

In Regions B and C of Figure 2, not only has the video release time become strictly positive, but it also continuously decreases toward a day-and-date release strategy. This result has an important empirical implication. Due to reasonably high variability on the many dimensions that characterize films, we can expect that γ_m varies significantly film by film. As the movie industry moves toward having more day-and-date releases, our model suggests that we should see video release times spanning the feasible range. As Disney CEO Robert Iger commented, “we’re not doing a one-size-fits-all approach” (Smith and Schuker 2010); we should therefore not expect videos to continue to be released according to an almost binary standard: either immediately or 3–4 months later. Instead, we anticipate seeing a more complete timespan utilized due to the diversity of film characteristics. This is a testable implication of our model.

As technology rapidly evolves, implementation in theater equipment often precedes consumer electronics. For example, the last few years have seen a rebirth of 3-D theatrical releases driven by improvements in 3-D technology, as seen with *Avatar*, *Alice in Wonderland*, and *Clash of the Titans*. However, 3-D televisions have just recently become available and have not yet achieved widespread adoption (Bonnington 2012). Thus, the relative quality of theatrical and video offerings can vary over time. People commonly believe that low theatrical quality films are the ones that should be released earlier to video (Epstein 2005). This intuition is often associated with the

observation that very low quality films sometimes bypass theatrical release entirely and appear directly on home video as presented in Proposition 2 and seen in Region A in Figure 2. Although it seems reasonable that higher quality movies are more likely to have longer release windows, this conclusion is not always justified. One important implication of our model is that an increase in the inherent quality of a movie, e.g., due to the 3-D theatrical releases, should sometimes be coupled with an *earlier* release time as illustrated in Regions B, C, and D in Figure 2, and formally established in Propositions 3 and 5. Hence, the optimal video release time is actually nonmonotonic in the inherent quality of the theatrical version of the film.

In contrast to the region studied above where the congestion parameter is high, we next investigate situations in which capacity constraints play a lesser role. For instance, the shadow price of capacity is likely to be lower in winter and spring in comparison to summer and holiday seasons (Einav 2007). Other factors that may decrease the congestion parameter include how the number of screens has increased over time, as well as possible local changes in the number of seats per screen (NATO 2013). Recently, small theater chains such as Cinapolis have competed by focusing on a higher quality experience through offering luxury seating, alcoholic beverages, full-service cafes, and an extensive selection of menu options for dining (Abate 2012, Luna 2012). In such cases, although there are fewer seats per screen, every seat is a “good seat” with unobstructed views, adequate spacing, and the ability to recline. As in these examples, if the congestion cost parameter becomes smaller, even the more congestion-sensitive class H consumers may now optimally choose to consume the theatrical movie offering.

Proposition 6 (Reduced Congestion Costs) *Suppose the quality-to-price ratio of the theatrical offering is sufficiently high and the congestion cost parameter is sufficiently low. Then,*

- (i) *If the film’s content durability is high, the studio releases the video at $T^* = \xi$ to induce multiple purchasing behavior and sets price to $p_d^* = \frac{\delta\gamma_d(1-\beta\xi)}{2}$;*
- (ii) *If the film’s content durability is low and the studio’s share of video revenue is larger than its share of movie revenue, i.e., $\lambda_m < \lambda_d$, the studio finds it profitable to pursue a day-and-date strategy by releasing the video immediately ($T^* = 0$) and pricing it at $p_d^* = \frac{\gamma_d(p_m(\lambda_m + \lambda_d) + \alpha\lambda_d(1-\rho))}{2\lambda_d(\gamma_m + \alpha(1-\rho))}$.*

When the congestion cost parameter is low, the more congestion-sensitive class H consumers have increased incentives to go to theaters. In particular, the consumers with higher quality sensitivity (i.e., sufficiently high $v \in \mathcal{V}$) will prefer to consume either both the movie and the video or just the movie in equilibrium (i.e., $\sigma^*(v, H) \in \{B, M\}$). When a film’s content durability is high, the studio will optimally adjust the video release time and price to provide stronger incentives for both classes of consumers to have the *both* segment represented in equilibrium. As stated in part (i) of Proposition 6, the studio optimally releases the video at $T^* = \xi$ to ensure that consumers choosing an equilibrium strategy of B will maintain maximum willingness to pay. The studio chooses this release time in conjunction with setting the monopoly price, $p_d^* = \frac{\delta\gamma_d(1-\beta\xi)}{2}$, for the residual

value associated with the durability of the video. Taken together, this delayed release strategy and optimal price ensure that no consumer will find it preferable to consume the video only (i.e., $\sigma^*(v, c) \neq D$ for all $v \in \mathcal{V}$); therefore, the consumer market equilibrium is fully characterized by (11) for both classes.

On the other hand, when the content durability of a film is low, the studio may instead find it optimal to pursue a day-and-date strategy. In particular, part (ii) of Proposition 6 establishes that when congestion costs are low and the quality-to-price ratio of the theatrical offering is high, the studio will hasten release of its video content whenever its share of video revenues is greater than its share of movie revenues. Lower content durability essentially leads to a shift in the studio's strategy. Because it is no longer concerned with preventing cannibalization of its *both* segment, there is less pressure to delay video release. Instead, the studio can release the video immediately and set price such that only class H consumers compose the *video* segment in equilibrium. Specifically, the video release time and price leads to a consumer market equilibrium where

$$\sigma^*(v, H) = \begin{cases} M & \text{if } \frac{\gamma_m(p_m - p_d) + \alpha(\gamma_m - \rho p_m)}{\gamma_m(\gamma_m - \gamma_d(1 - \beta T) + \alpha(1 - \rho))} \leq v < 1; \\ D & \text{if } \frac{p_d}{\gamma_d(1 - \beta T)} \leq v < \frac{\gamma_m(p_m - p_d) + \alpha(\gamma_m - \rho p_m)}{\gamma_m(\gamma_m - \gamma_d(1 - \beta T) + \alpha(1 - \rho))}; \\ N & \text{if } v < \frac{p_d}{\gamma_d(1 - \beta T)}, \end{cases} \quad (14)$$

and $\sigma^*(v, H)$ also satisfies (14) with $\alpha = 0$. In this case, when the studio's share of revenues for the video increases, the studio decreases its video price in order to expand the video market, whereas when the studio's share of revenues for the theatrical movie offering increases, the studio increases its video price.

Having seen how varying levels of congestion costs can lead to diverse studio strategies and their associated consumer market structures in equilibrium, we now turn attention to how congestion impacts profitability. On the surface, one may think that an increase in the congestion cost parameter *always* leads to lower profits because it directly hurts the utility of class H consumers. However, in the following proposition, we establish that, in some cases, a higher congestion cost parameter can actually benefit the studio.

Proposition 7 (Congestion and Profitability) *If the quality-to-price ratio of the theatrical offering is sufficiently high and the content durability is sufficiently low, an increase in the congestion cost parameter increases the studio's profit whenever $\lambda_d > \lambda_m \max(1, 4p_m/\gamma_d)$ is satisfied.*

Proposition 7 has an interesting message: congestion can sometimes increase the profitability of a film. In particular, when the studio's revenue share from video sales sufficiently exceeds its share from movie sales, the studio finds the *video* market quite appealing because of its higher margins. However, the quality-to-price ratio (γ_m/p_m) of the theatrical offering is quite high and thus will provide moviegoers with a lot of surplus. Consumer surplus is especially high for both classes of consumers when the congestion cost parameter is low. In such an instance, it becomes difficult for the studio to create incentives for consumers to purchase the video without substantially reducing

the price p_d due to the high value obtainable in theaters. However, when the congestion cost parameter becomes high, the congestion-sensitive consumers in class H react to the externality imposed on them by class L consumers. In fact, with high congestion costs, these consumers will have much stronger incentives to consume the video instead of the movie. Because of these inherent incentives, the studio can now perfectly segment the consumer market and charge the class L consumers a price reflecting monopoly power over content yielding substantial profits because of its favorable revenue share. Essentially, the studio leverages congestion as a tool to separate the market and charge higher prices.

In the previous proposition, we studied how the studio can harness variation in consumer heterogeneity with regard to congestion toward profitability. We next develop a better understanding of how the studio's release strategy changes as the extent of heterogeneity in the consumer population decreases, e.g., a film's target audience becomes homogeneous.

Proposition 8 (Consumer Heterogeneity) *If the quality-to-price ratio of the theatrical offering and the congestion parameter are sufficiently high and content durability is sufficiently low, then a studio responds to an increase in ρ by switching from a perfect segmentation strategy, i.e., $T^* = 0$ and $p_d^* = \frac{\gamma_d}{2}$ for low ρ , to a delayed release strategy with $T^* = \xi$ and $p_d^* = \frac{p_m \gamma_d (1 - \beta \xi)}{\gamma_m}$ to induce multiple purchasing behavior for high ρ .*

A moviegoer population characterized by more class L consumers (larger ρ) corresponds to essentially a more homogeneous, congestion-insensitive audience. Studios producing films that cater to such an audience may find it profitable to pursue a strategy that induces a significant *both* consumer segment in equilibrium. For example, films such as *Justin Bieber: Never Say Never* and *Twilight* which target the teenage girl and younger women demographic would benefit from utilizing a finely tuned *delayed released* strategy to profit from incentive compatible multiple purchases (Marks 2008, Itzkoff 2011). However, when the targeted demographic is characterized by a greater proportion of more congestion-sensitive moviegoers, a studio is better off releasing the video immediately and segmenting the market by classes as part of a day-and-date strategy. Because of the increased number of consumers in class H , the studio prefers to price the video higher and forgo multiple purchasing behavior.

6 Piracy, Pricing Constraints, and Welfare

Piracy is a pervasive problem in digital goods markets and is often facilitated by online systems that support illegal filesharing. Researchers who study the economics of digital piracy identify how consumers, when they pirate, often still face costs that are akin to price. For example, pirates are subject to government imposed penalties, search and learning costs, and moral costs (Connor and Rumelt 1991, August and Tunca 2008, Danaher et al. 2010). Because would-be pirates are an important source of revenue to studios, the more general pricing problem they face must also account for the incentive compatibility constraints of consumers considering piracy. To provide

insight into how movie piracy would affect the studio's setting of video release time and price controls, in this section we formally examine the studio's behavior as its pricing problem becomes more constrained as a consequence of combatting piracy (Pogue 2012).

In the following proposition, we examine how a price constraint affects the studio's profit-maximization problem. If the quality of the pirated and actual product are on par, then deterrence is equivalent to a constraint on price.¹³

Proposition 9 (Content Piracy) *Under a constraint on the video price, i.e., $p_d \leq c_d$ with small c_d , suppose the quality-to-price ratio of the theatrical offering is intermediate, the congestion parameter is high, content durability is sufficiently low, and $\lambda_d > \rho\lambda_m$. Then,*

(i) *If γ_d is small (but satisfies $\gamma_d > c_d$), then p_d^* is interior, i.e., $p_d^* = \frac{\gamma_d(\lambda_d(1-\rho)(\gamma_m-\gamma_d)+\rho p_m(\lambda_m+\lambda_d))}{2\lambda_d(\gamma_m-(1-\rho)\gamma_d)} < c_d$ and $T^* = 0$;*

(ii) *If γ_d is big, then $p_d^* = c_d$ and $T^* > 0$ and satisfies:*

$$T^* = \inf \{T \mid \rho\gamma_d^2(p_m - c_d)(\lambda_m p_m - \lambda_d c_d)(1 - \beta T)^2 - \lambda_d c_d^2(\gamma_m - \gamma_d(1 - \beta T))^2 = 0\}. \quad (15)$$

Moreover, when c_d becomes sufficiently small, T^ approaches $\frac{1}{\beta} \left(1 - \frac{\gamma_m c_d}{p_m \gamma_d} \sqrt{\frac{\lambda_d}{\rho \lambda_m}}\right)$.*

To provide a clearer discussion of the impact of piracy, we focus our analysis on a parameter region where the *both* consumer market segment does not arise in equilibrium; hence, whether to attribute changes in video release timing and pricing to provision of incentives for multiple purchases becomes irrelevant. Instead, under the conditions of Proposition 9, the equilibrium consumer market structure has *movie*, *video* and *none* segments for class *L* consumers while only *video* and *none* segments are represented for class *H* consumers.

Proposition 9 establishes that the quality of a studio's video offering determines the extent to which constrained pricing due to piracy affects the studio's release strategy. Currently, a few popular video options for film content are DVD, Blu-ray Disc (BD) and video-on-demand (VOD); further, VOD is becoming an increasingly viable channel that can be disruptive to long-standing business models (Zhu 2001, Sullivan 2012). Comparing DVD/BD to VOD, and all else being equal, one may argue that BD and DVD are higher quality offerings than VOD, i.e., they have a higher associated γ_d . For example, a film offered over VOD may be viewable for only 24-48 hours, may be influenced by consumer bandwidth limitations, and may not carry any of the additional content often found on DVDs and BDs. Part (i) of Proposition 9 suggests that a studio intending to release a lower quality, VOD offering after theatrical release should actually release it immediately at $T^* = 0$ and reduce its price as part of a day-and-date strategy. With a lower quality, VOD offering, the price a studio sets is also lower and largely driven by class *H* consumers who will not consume

¹³One can formally model the piracy option in the consumer's decision problem as in Connor and Rumelt (1991) and August and Tunca (2008) to see this equivalence.

the movie offering. In this case, piracy will be effectively deterred by the lower prices and make releasing the VOD immediately an optimal strategy.

Recently, we have seen studios effectively deploying this strategy. *Margin Call* is an Indie film about the 2008 financial crisis that was released in 2011 with a day-and-date strategy using the VOD channel. The VOD offering was priced lower at \$6.99 and was fairly successful generating VOD revenues of \$8 million which exceeded its domestic box office receipts of \$5.3 million (Snyder 2011, Miller 2012, McClintock 2013). Lionsgate, the film’s distributor, successfully executed another day-and-date VOD strategy for the financial drama *Arbitrage* in 2012 with VOD revenues of \$14 million and a domestic box office of \$8 million (McClintock 2013). In a similar vein, in 2004 the Convex Group released *Noel* using a day-and-date strategy with disposable DVDs. Like the VOD offerings, disposable DVDs have a time limitation (e.g., 24-48 hours) which reduces γ_d . Consistent with part (i) of Proposition 9, the Convex Group released the DVD simultaneously with theatrical release and for a reduced price of \$4.99 (Abramowitz 2004).

On the other hand, if the quality of the video offering is higher (i.e., higher γ_d), the studio may instead find it optimal to pursue a delayed release strategy. As γ_d becomes larger, the studio increases p_d strategically to achieve two purposes: (i) accrue higher revenues from the class H consumers, and (ii) limit cannibalization of its *movie* consumer market segment among class L consumers. However, the studio’s incentive to strategically raise price runs counter to its other objective to limit piracy. Part (ii) of Proposition 9 establishes that when its video price is lowered to deter piracy, the studio finds it preferable to delay the video release. In this case, piracy concerns drive the studio to use delayed release as an alternative control mechanism to help limit cannibalization of movie revenues.

In practice, as referenced earlier, the average DVD release window is 103.4 days thus far in 2013. Piracy has long been a major concern of the Motion Picture Association of America (MPAA) and the National Association of Theater Owners (NATO), and the historically delayed video release policy has been a partially effective deterrent (Cieply 2011). However, with an increasing consumer appetite for immediate content availability, studios are looking for other ways to shorten the DVD release window while still addressing piracy. Our results suggest that offering plain vanilla, lower-quality versions of film content (i.e., without alternate endings, deleted scenes and other special features) at a lower price would be an effective strategy that goes hand-in-hand with an earlier video release. Higher quality, full-fledged DVDs and BDs can follow later with all of the special features at a higher price. In addition to being effective for piracy control, such a strategy may also mesh well with supply side constraints since higher quality versions take longer to produce (Cole 2007).

In Figure 3, we illustrate how the studio adapts its video release time strategy over a wide range of values for content durability (δ) and the congestion cost parameter (α) when it prices to deter piracy. As we established in Proposition 4 and part (ii) of Proposition 6, the studio should pursue a day-and-date strategy, possibly achieving perfect segmentation, when content durability

Optimal Video Release Time

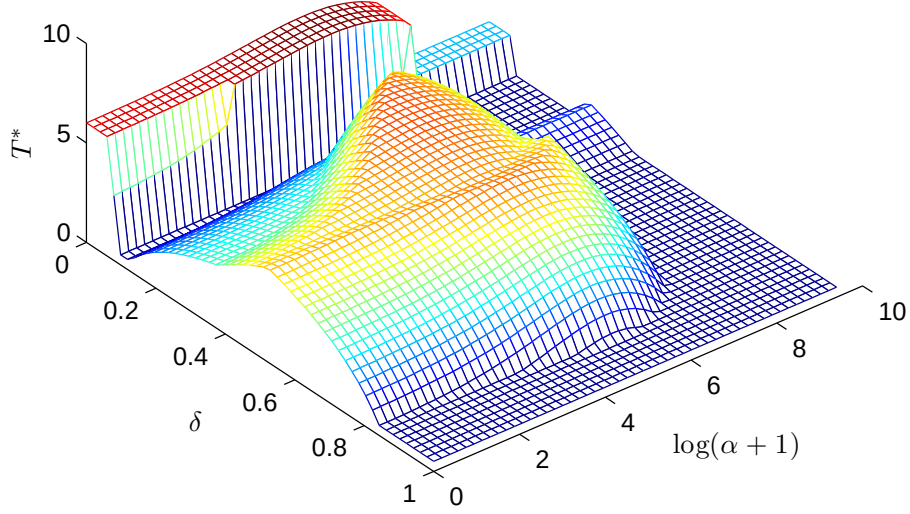


Figure 3: How the studio's optimal release time is affected by congestion externalities and film content durability under piracy deterrence. The parameter values are $\beta = 0.1$, $\gamma_m = 240$, $\gamma_d = 80$, $\lambda_m = 0.4$, $\lambda_d = 0.7$, $p_m = 50$, $c_d = 5$, $\rho = 0.2$, and $\xi = 0.5$.

is low. In both cases, the studio optimally sets $T^* = 0$. In contrast, with piracy deterrent pricing, Figure 3 reflects that the studio is forced to delay release as indicated in part (ii) of Proposition 9. Specifically, examining the region under low δ and high α , Figure 3 illustrates how the studio shifts out T^* as part of its delayed release strategy. The optimal release time surface also exhibits other interesting aspects. For example, examining any cross-section in α , as δ increases from approximately 0.05 to 0.1, the studio's optimal video release time jumps down from a high T^* (anywhere from 2.3 to 6.7) to a much earlier time, $T^* = 0$. The studio's behavior has significant welfare implications because delayed video releases, which go hand-in-hand with decayed quality, tend to hurt consumer surplus. Hence, policy can play an important role. For example, social planners may consider subsidizing day-and-date releases or strategically taxing/subsidizing video purchases to expand the region of lower film durability in which studios find it optimal to pursue day-and-date strategies.

Surprisingly, social welfare can also be hurt by increases in the inherent quality of the video offering which we formally establish in the following proposition.

Proposition 10 *An increase in the quality of the video can result in welfare losses.*

The driver of this finding is the strategic behavior of the studio as it attempts to maximize profitability when it is concerned about piracy. As we saw in part (i) of Proposition 9, when γ_d is not too big, a day-and-date strategy is optimal. Even when day-and-date strategies are not profit-maximizing, they can enhance social welfare. Because these strategies make both theatrical and video versions of the film available early, consumer surplus can be much higher because (i) more

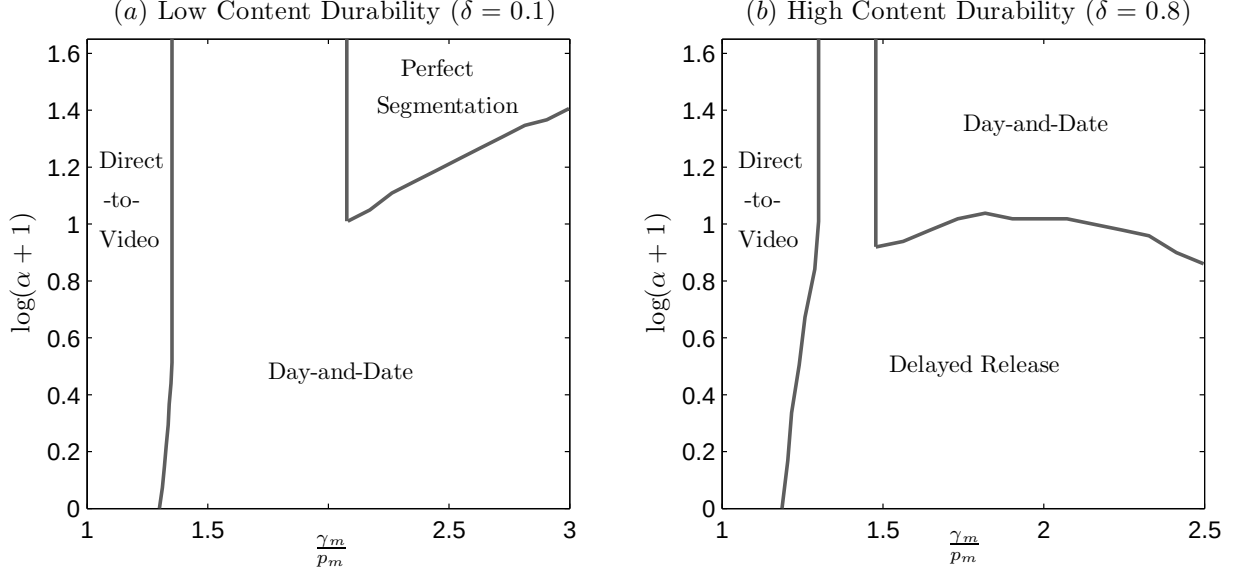


Figure 4: Studio's optimal video release strategies depending on movie characteristics. Parameter values are $\beta = 0.2$, $\gamma_d = 0.4$, $\lambda_m = 0.5$, $\lambda_d = 0.6$, $p_m = 0.5$, $\rho = 0.4$ and $\xi = 1.25$ for both panels.

consumers make multiple purchases, (ii) consumers who would have purchased the video anyway can now purchase it earlier, before its quality decays, and (iii) consumers who were left out of the market can now consume the video and obtain positive surplus. From part (ii) of Proposition 9, we saw that when γ_d is at a higher level, the studio must use its video release time strategically to inhibit cannibalization by decaying the quality of the video through a delayed release strategy. This behavior, although profit-maximizing, hurts consumer surplus substantially and can lead to an overall welfare decrease as stated in Proposition 10.

7 Discussion and Concluding Remarks

In this paper, we present a model of film distribution and consumption to gain insight into how studios should optimally price and time the release of video versions of their films when accounting for strategic behavior of consumers in product choice. We take a normative approach to the studio's problem and highlight the critical factors that can motivate direct-to-video, day-and-date, perfect segmentation, and delayed release strategies. In developing the consumer model, we incorporate several important factors: (i) quality decay of the film content over time; (ii) the quality/price gap between theater and video alternatives; (iii) consumption of both the theatrical and video version and quantification of these preferences using a notion of content durability (including delayed value realization for consumers who purchase both); and (iv) either negative consumption externalities associated with congestion at theaters or the absence of related costs for heterogeneous classes of consumers. These factors, which are in turn influenced by video release timing and pricing, together affect consumption behavior.

Figure 4 provides a visual summary of our findings. We analyze a variety of dimensions, including the quality-to-price ratio of the theatrical offering, how durable the film’s content is in terms of repeat viewings, and whether congestion in theaters can be effectively controlled. Panels (a) and (b) correspond to the low content durability and high content durability cases, respectively. We demonstrate that for films with lower movie quality-to-price ratios, a direct-to-video strategy maximizes studio profits. When content durability is high as in panel (b), as the quality-to-price ratio becomes higher, the studio finds it preferable to switch to a delayed video release strategy and a correspondingly lower price in order to induce less congestion sensitive consumers to both attend theaters and purchase videos. As the ratio becomes even larger, the studio shifts to a day-and-date strategy provided congestion costs are not too low, in which case cannibalization of movie revenues becomes less of an issue. On the other hand, when content durability is low, as in panel (a), the studio will adjust the video price upward because providing the incentives necessary to induce a segment of consumers who choose both offerings is unprofitable, and achieve perfect segmentation. However, when congestion costs are low and the studio’s share of video revenues are favorable, a day-and-date strategy is again optimal for the studio as it adjusts the video price to expand the video market to less congestion-sensitive consumers.

Using our model, we analytically established many propositions with a wide range of relevant results to studios. These results have many testable implications.

1. For the mid-to-high portion of the quality space, as theatrical movie quality increases, the video release time decreases.
2. For the low portion of the quality space, as a film’s theatrical movie quality decreases, it is more likely to be released using a direct-to-video strategy.
3. For high quality movies with high content durability, as congestion in theater increases, e.g., due to peak season capacity constraints, more day-and-date strategies will be implemented.
4. When congestion effects are low, films with higher content durability are more likely to be released later; however, films with lower content durability should be released using a day-and-date strategy if the studio’s revenue share of the video channel is more attractive than the theater channel.
5. For films with relatively homogeneous audiences that are not sensitive to congestion, we expect to see studios using delayed release strategies with prices that induce consumers to purchase both movie and video offerings.
6. As piracy becomes more severe, more lower quality videos (including VOD and disposable DVDs) will be released sooner with lower prices, followed by higher quality videos (including full-fledged DVDs and BD) released later with higher prices.

Many of these testable implications require parameters such as content durability to be measured. Such parameters can be forecasted from data on consumer viewing habits. For example, data from Nielsen can help determine how durable different genres of film are as well as how content durability might relate to a film’s target audience. We ran an initial survey that showed the *action* genre is the most durable, and the *documentary* genre is the least durable. The other parameters that constitute our model can similarly be measured using appropriate data sets.

A number of important questions remain for future research. For example, in this work, we focused our attention on the optimal video release timing and pricing strategy for a single movie. Studying a single movie setting is important since, in many cases, studios have some degree of freedom in the timing of video releases; in addition, such a model more readily clarifies the central trade-offs. However, competition is certainly an important topic and can even be a factor for deciding when to schedule theatrical release, particularly during peak seasons; but, the effect of competition on *video* release times is much weaker (Goldberg 1991). Consequently, the insights we provide in this work should remain valid for most movies. Another aspect that remains to be investigated is the release of multiple films by the same studio over time, i.e., the repeated game aspect. Along these lines, Prasad et al. (2004) specifically consider the impact of consumer expectations over time in an aggregate model. In that context, the authors commented that an interesting avenue of future research would be to start from a consumer utility model and capture the impact of consumer expectations which arise endogenously. Our paper may serve as a building block for further research in this direction.

In this paper, we study the video release problem in the context of a fixed simple revenue sharing contract between studios and exhibitors. In practice, conventional contracts involve time-dependent revenue sharing (i.e., a sliding, increasing percentage of revenues to exhibitors) after allowance for exhibitors’ expenses. But recently, some studios and exhibitors have begun to implement “aggregate settlement,” a simple revenue split without sliding scales (Vogel 2007). Our model can serve as a reasonable approximation for that scenario. However, we have made simplifying assumptions in order to focus primarily on the underlying trade-offs relevant to our research questions. Nevertheless, our results are robust and satisfied for wide ranges of revenue shares. Looking ahead, we believe one fruitful extension would be to rigorously analyze how the various contracts between studios, theater owners, and video retailers should be designed, particularly in light of our results on the profitability of day-and-date release strategies which are not yet widely employed in industry.

We assumed here that movie quality is certain and common knowledge, noting that empirical evidence concerning demand uncertainty associated with quality is not as strong as popularly argued (see, e.g., Orbach 2004, Orbach and Einav 2007). Furthermore, most of the uncertainty is revealed after the first weekend of release. Consequently, when distributors set video release dates after uncertainties are almost resolved and consumers are well-informed, most of our results are preserved. If consumers are not informed about the quality of films, then video windows determined

by distributors may signal quality, a scenario which could be a useful extension of our work. In this direction, our paper provides an interesting observation: Contrary to conventional wisdom, longer video release windows do not necessarily signal higher movie quality in a context of multiple purchases.

Finally, in our study, we assumed that the studio announces the video release time and price immediately and commits to both which seems credible given the repeated nature of the context. Commitment is a non-issue when either day-and-date or direct-to-video strategies are optimal since the video channel is opened immediately. However, a studio's inability to announce immediately can affect consumption early on during the run of a film in theaters. Extending the model to permit slightly delayed announcements could be worth studying, especially for cases where video release times are optimally determined to be less than a month out.

This paper is a first step toward analyzing the trade-offs faced by studios as they determine an appropriate video release time and pricing strategy. Studios are demonstrably interested in the prospects of earlier release times and even day-and-date strategies, and can benefit from a better understanding of how congestion, film content durability, and piracy interact with such strategies. Given the speed of technological advances and the enduring use of a channeling system, the film industry now has a significant opportunity to design products and make decisions on delivery systems that cater more effectively to consumer preferences. By establishing how effective day-and-date strategies are at boosting studio profits, we hope that our work helps to initiate part of this progress.

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Appendix for “Optimal Timing of Sequential Distribution: The Impact of Congestion Externalities and Day-and-Date Strategies”

A Proofs of Propositions and Lemmas

Proof of Proposition 1 We first consider class $c \in L$. In this class of customers, the required conditions for $\sigma^*(v, L) = B$ are the following three:

$$V(v, L, B, \sigma_{-v}^*) \geq V(v, L, M, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_d}{\delta\gamma_d(1 - \beta \max(T, \xi))}; \quad (16)$$

$$V(v, L, B, \sigma_{-v}^*) \geq V(v, L, D, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \max(T, \xi)))}; \quad (17)$$

$$V(v, L, B, \sigma_{-v}^*) \geq V(v, L, N, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_m + p_d}{\gamma_m + \delta\gamma_d(1 - \beta \max(T, \xi))}. \quad (18)$$

As a result, it follows that $\sigma^*(v, L) = B$ if and only if $v \geq \omega_b^L$, where

$$\omega_b^L \triangleq \max \left(\frac{p_d}{\delta\gamma_d(1 - \beta \max(T, \xi))}, \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \max(T, \xi)))}, \frac{p_m + p_d}{\gamma_m + \delta\gamma_d(1 - \beta \max(T, \xi))} \right). \quad (19)$$

Next, for $\sigma^*(v, L) = M$ consumer strategy, we need the following three conditions:

$$V(v, L, M, \sigma_{-v}^*) \geq V(v, L, B, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_d}{\delta\gamma_d(1 - \beta \max(T, \xi))}; \quad (20)$$

$$V(v, L, M, \sigma_{-v}^*) \geq V(v, L, D, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}; \quad (21)$$

$$V(v, L, M, \sigma_{-v}^*) \geq V(v, L, N, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_m}{\gamma_m}. \quad (22)$$

As a result, $\sigma^*(v, L) = M$ if and only if

$$\max \left(\frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}, \frac{p_m}{\gamma_m} \right) \leq v \leq \frac{p_d}{\delta\gamma_d(1 - \beta \max(T, \xi))}. \quad (23)$$

Note that $\frac{p_d}{\delta\gamma_d(1 - \beta \max(T, \xi))} \leq \omega_b^L$. Third, the following three conditions are required in order to have $\sigma^*(v, L) = D$:

$$V(v, L, D, \sigma_{-v}^*) \geq V(v, L, B, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \max(T, \xi)))}; \quad (24)$$

$$V(v, L, D, \sigma_{-v}^*) \geq V(v, L, M, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}; \quad (25)$$

$$V(v, L, D, \sigma_{-v}^*) \geq V(v, L, N, \sigma_{-v}^*) \Leftrightarrow v \geq \frac{p_d}{\gamma_m(1 - \beta T)}. \quad (26)$$

Consequently, $\sigma^*(v, L) = D$ if and only if

$$\frac{p_d}{\gamma_m(1-\beta T)} \leq v \leq \min \left(\frac{p_m}{\gamma_m - \gamma_d(1-\beta T - \delta(1-\beta \max(T, \xi)))}, \frac{p_m - p_d}{\gamma_m - \gamma_d(1-\beta T)} \right). \quad (27)$$

In addition, it follows that

$$\min \left(\frac{p_m}{\gamma_m - \gamma_d(1-\beta T - \delta(1-\beta \max(T, \xi)))}, \frac{p_m - p_d}{\gamma_m - \gamma_d(1-\beta T)} \right) \leq \max \left(\frac{p_m - p_d}{\gamma_m - \gamma_d(1-\beta T)}, \frac{p_m}{\gamma_m} \right). \quad (28)$$

Lastly, the required conditions for $\sigma^*(v, L) = N$ are the following three:

$$V(v, L, N, \sigma_{-v}^*) \geq V(v, L, B, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_m + p_d}{\gamma_m + \delta\gamma_d(1-\beta \max(T, \xi))}; \quad (29)$$

$$V(v, L, N, \sigma_{-v}^*) \geq V(v, L, M, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_m}{\gamma_m}; \quad (30)$$

$$V(v, L, N, \sigma_{-v}^*) \geq V(v, L, D, \sigma_{-v}^*) \Leftrightarrow v \leq \frac{p_d}{\gamma_m(1-\beta T)}. \quad (31)$$

As a result, it follows that $\sigma^*(v, L) = N$ if and only if $v \leq \omega_d^L$, where

$$\omega_d^L \triangleq \min \left(\frac{p_m + p_d}{\gamma_m + \delta\gamma_d(1-\beta \max(T, \xi))}, \frac{p_m}{\gamma_m}, \frac{p_d}{\gamma_m(1-\beta T)} \right). \quad (32)$$

Moreover, we have $\omega_d^L \leq \frac{p_d}{\gamma_m(1-\beta T)}$. Using these characterizations and relationships among boundaries together with $\sigma^*(v, L) \in \{N, D, M, B\}$ for all $v \in \mathcal{V}$, (4) follows. Note that depending on the parameter values, it is possible that some strategies may not be in equilibrium, e.g., if $\omega_m^L > \omega_b^L$, then $\sigma^*(v, L) \neq M$ for all $v \in \mathcal{V}$. Now, consider the class H consumers. The only difference in this class is that for both $V(v, H, B, \sigma_{-v}^*)$ and $V(v, H, M, \sigma_{-v}^*)$, there exists an additional congestion cost, $\alpha\eta$, which is fixed given the other consumers' equilibrium strategies, i.e., σ_{-v}^* . Closely following the previous arguments for the L class consumers with this difference, we also obtain (4). This completes the proof. ■

Proof of Proposition 2 Technically, we will prove that there exist $\omega_1 > 0$ such that if $\frac{\gamma_m}{p_m} < \omega_1$, then $T^* = 0$ and $p_d^* = \frac{\gamma_d}{2}$.

First, suppose that $\sigma^*(1, L) \in \{B, M\}$ in equilibrium. Then, by (2), it must follow that $\gamma_m - p_m > \gamma_d(1-\beta \max(T, \xi)) - p_d$. Dividing by p_m and letting $\frac{\gamma_m}{p_m} = 1 + K_1\zeta$, we obtain

$$\frac{\gamma_d(1-\beta \max(T, \xi)) - p_d}{p_m} < \frac{\gamma_m}{p_m} - 1 = K_1\zeta. \quad (33)$$

By (33), p_d must satisfy $p_d > \gamma_d(1 - \beta \max(T, \xi)) - p_m K_1 \zeta$. Thus, by (1), (2), (5) and (6), $D_m \leq 1 - \frac{p_m}{\gamma_m} = 1 - \frac{1}{1+K_1\zeta}$ and $D_d \leq 1 - \frac{p_d}{\gamma_d(1-\beta \max(T, \xi))} < \frac{p_m K_1 \zeta}{\gamma_d(1-\beta \max(T, \xi))}$. By (7),

$$\begin{aligned} \Pi(T, p_d) &= \lambda_m p_m D_m + \lambda_d p_d D_d \\ &< \lambda_m p_m \left(1 - \frac{1}{1+K_1\zeta}\right) + \lambda_d p_d \left(\frac{p_m K_1 \zeta}{\gamma_d(1-\beta \max(T, \xi))}\right). \end{aligned} \quad (34)$$

Second, suppose that $\sigma^*(1, L) \in \{D\}$ in equilibrium. Then, by Proposition 1, $\sigma^*(v, L) \in \{D, N\}$ and $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$, and by (5), (6), and (7), we obtain

$$\Pi(T, p_d) = \lambda_d p_d \left(1 - \frac{p_d}{\gamma_d(1 - \beta \max(T, \xi))}\right). \quad (35)$$

By (8) and optimizing (35), we obtain maximizers $\hat{p}_d = \frac{\gamma_d}{2}$ and $\hat{T} = 0$, hence $\Pi(\hat{T}, \hat{p}_d) = \frac{\lambda_d \gamma_d}{4}$. Comparing this expression to the upper bound on profits from (34), for sufficiently small ζ , it follows that $p_d^* = \hat{p}_d$ and $T^* = \hat{T}$. ■

Proof of Proposition 3 Technically, we will prove that there exists $\omega_2 > 0$ such that if $\frac{\gamma_m}{p_m} > \omega_2$, $\alpha > \omega_2$, and $\delta > 1 - \frac{1}{\omega_2}$, then $T^* = 0$ and $p_d^* = \frac{\delta \gamma_d(1-\beta\xi)}{2(\delta(1-\beta\xi)(1-\rho)+\rho)}$.

We first prove the following Lemma, which will be used for the proof of this proposition as well as the proof of Proposition 5 as well as the other following proof of propositions which require high α values:

Lemma 1 *If $\alpha > \frac{\gamma_m}{\rho}$, then $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$.*

Proof: First, suppose that $\sigma^*(v, L) \in \{D, N\}$ for all $v \in \mathcal{V}$. From $V(v, H, s, \sigma_{-v}^*) \leq V(v, L, s, \sigma_{-v}^*)$ for $s \in \{M, B\}$, for all $v \in \mathcal{V}$, and $V(v, H, s, \sigma_{-v}^*) = V(v, L, s, \sigma_{-v}^*)$ for $s \in \{D, N\}$, for all $v \in \mathcal{V}$, it follows that $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$.

Second, suppose that $\sigma^*(v, L) \in \{B, M\}$ for some $v \in \mathcal{V}$. Denote the market structure (*none, video, movie, both*) as the one in (4) with $0 < \omega_d^c < \omega_m^c < \omega_b^c < 1$, i.e., all market segments have strictly positive market demand. Similarly, we also denote the market structure (*none, video, movie*) as the one in (4) with $0 < \omega_d^c < \omega_m^c < \omega_b^c = 1$. We then first consider the following equilibrium consumer market structures for class L : (*none, video, movie, both*) or (*none, video, movie*). In these two structures, it follows that $\eta \geq \rho(1 - \frac{p_m - p_d^*}{\gamma_m - \gamma_b(1 - \beta T^*)})$. We then obtain

$$\begin{aligned} V(v, H, M, \sigma_{-v}^*) = \gamma_m v - \alpha \eta - p_m &\leq \gamma_m v - \alpha \rho \left(1 - \frac{p_m - p_d^*}{\gamma_m - \gamma_b(1 - \beta T^*)}\right) - p_m \\ &< \gamma_m v - \gamma_m \left(1 - \frac{p_m - p_d^*}{\gamma_m - \gamma_b(1 - \beta T^*)}\right) - p_m \\ &< \gamma_d(1 - \beta T^*)v - p_d^* = V(v, H, D, \sigma_{-v}^*), \end{aligned} \quad (36)$$

for all $v \in \mathcal{V}$, in which the second inequality follows from $\alpha > \frac{\gamma_m}{\rho}$ and the last inequality follows from $\frac{p_m - p_d^*}{\gamma_m - \gamma_b(1 - \beta T^*)} < 1$. Hence, $V(v, H, M, \sigma_{-v}^*) < V(v, H, D, \sigma_{-v}^*)$ for all $v \in \mathcal{V}$. Next, if the market structure for class L is $(none, video, movie)$, then from $V(v, H, B, \sigma_{-v}^*) \leq V(v, L, B, \sigma_{-v}^*)$ for all $v \in \mathcal{V}$, there cannot be *both* segment in class H . Together with (36), in the case of $(none, video, movie)$ for class L , the only possible equilibrium market structure for class H is $(none, video)$. Next, if the market structure for class L is $(none, video, movie, both)$, then

$$\begin{aligned} V(v, H, B, \sigma_{-v}^*) &= \gamma_m v - \alpha \eta - p_m + \delta \gamma_d (1 - \beta \max(T^*, \xi)) v - p_d^* \\ &< \gamma_m v - \gamma_m \left(1 - \frac{p_m - p_d^*}{\gamma_m - \gamma_b(1 - \beta T^*)} \right) - p_m + \delta \gamma_d (1 - \beta \max(T^*, \xi)) v - p_d^* \\ &< \gamma_d (1 - \beta T^*) v - p_d^* = V(v, H, D, \sigma_{-v}^*), \end{aligned} \quad (37)$$

for all $v \in \mathcal{V}$, in which the last inequality is obtained using $V(1, L, B, \sigma_{-v}^*) > V(1, L, D, \sigma_{-v}^*)$ and the existence of strictly positive *movie* segment for class L , i.e., $\frac{p_m - p_d^*}{\gamma_m - \gamma_d(1 - \beta T^*)} < \frac{p_d^*}{\delta \gamma_d(1 - \beta \max(T^*, \xi))}$. As a result, in the case of $(none, video, movie, both)$ for class L , from (36) and (37), it follows that $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$. Similarly, for the other remaining consumer equilibrium market structures for class L , specifically, for the market structures of $(none, movie, both)$, $(none, movie)$, and $(none, video, both)$, we can also show $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$ from $\alpha > \frac{\gamma_m}{\rho}$. ■

For the proof of Proposition 3, because α is sufficiently high, we have $\sigma^*(v, H) \in \{D, N\}$ for all $v \in \mathcal{V}$ and the consumer equilibrium market structure is given in (10) for $c = H$, or all N , i.e., $\sigma^*(v, H) \in \{N\}$ for all $v \in \mathcal{V}$. For class L consumers, we first show that when δ is high and $T < \xi$, consumer equilibrium market structure is as follows:

$$\begin{cases} (none, video, both) & \text{if } 0 \leq p_d \leq \frac{p_m \gamma_d \delta (1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}; \\ (none, video, movie, both) & \text{if } \frac{p_m \gamma_d \delta (1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))} < p_d \leq \frac{p_m \gamma_d (1 - \beta T)}{\gamma_m}; \\ (none, movie, both) & \text{if } \frac{p_m \gamma_d (1 - \beta T)}{\gamma_m} < p_d \leq \gamma_d \delta (1 - \beta \xi); \\ (none, movie) & \text{if } \gamma_d \delta (1 - \beta \xi) < p_d. \end{cases} \quad (38)$$

Consider the first region, in which $p_d \leq \frac{p_m \gamma_d \delta (1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$. If consumers prefer *movie* over *both*, it implies that $V(v, L, M, \sigma_{-v}^*) \geq V(v, L, B, \sigma_{-v}^*)$, which is written as $v < \frac{p_d}{\delta \gamma_d(1 - \beta \xi)}$. Furthermore, if consumers prefer *movie* over *video*, i.e., $V(v, L, M, \sigma_{-v}^*) \geq V(v, L, D, \sigma_{-v}^*)$, we can rewrite this condition as $v > \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}$. However, under the given p_d condition, $p_d \leq \frac{p_m \gamma_d \delta (1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$, it follows that $\frac{p_d}{\delta \gamma_d(1 - \beta \xi)} < \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}$. Consequently, in this p_d region, $\sigma^*(v, L) \in \{B, D, N\}$, i.e., no consumer purchases theatrical movie offering only. Next, in order to have $\sigma^*(v, L) = B$, we need both of the following conditions to be satisfied: (i) $V(v, L, B, \sigma_{-v}^*) \geq V(v, L, D, \sigma_{-v}^*)$, which translates into $v \geq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$; (ii) $V(v, L, B, \sigma_{-v}^*) \geq V(v, L, N, \sigma_{-v}^*) = 0$, which is simplified to $v \geq \frac{p_m + p_d}{\gamma_m + \delta \gamma_d(1 - \beta \xi)}$. Using high δ , $T > \xi$, and the given p_d condition, we then obtain $\frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))} \geq \frac{p_m + p_d}{\gamma_m + \delta \gamma_d(1 - \beta \xi)}$. As a result, $\sigma^*(v, L) = B$ if and only if $v \geq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$.

Next, the required conditions for $\sigma^*(v, L) = D$ are: (i) $V(v, L, D, \sigma_{-v}^*) \geq V(v, L, B, \sigma_{-v}^*)$, which is equivalent to $v \leq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$; (ii) $V(v, L, D, \sigma_{-v}^*) \geq V(v, L, N, \sigma_{-v}^*) = 0$, which can be rewritten as $v \geq \frac{p_d}{\gamma_d(1 - \beta T)}$. Moreover, under the given conditions, it follows that $\frac{p_d}{\gamma_d(1 - \beta T)} < \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$. Hence, we have $\sigma^*(v, L) = D$ if and only if $\frac{p_d}{\gamma_d(1 - \beta T)} \leq v \leq \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$. In summary, when $p_d \leq \frac{p_m \gamma_d \delta(1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}$, the consumer consumption equilibrium strategy for class L is

$$\sigma^*(v, L) = \begin{cases} B & \text{if } \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))} \leq v \leq 1; \\ D & \text{if } \frac{p_d}{\gamma_d(1 - \beta T)} \leq v < \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}; \\ N & \text{if } v < \frac{p_d}{\gamma_d(1 - \beta T)}. \end{cases} \quad (39)$$

In this region, the studio's profit function becomes

$$\Pi(T, p_d) = \lambda_m p_m \rho \left(1 - \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))} \right) + \lambda_d p_d \left(1 - \frac{p_d}{\gamma_d(1 - \beta T)} \right). \quad (40)$$

We first consider the optimal video pricing problem given T . Taking the derivative of $\Pi(T, p_d)$ in (40) with respect to p_d , we obtain

$$\frac{\partial \Pi}{\partial p_d} = \lambda_d \left(1 - \frac{2p_d}{\gamma_d(1 - \beta T)} \right), \quad (41)$$

which is satisfied at $p_d^*(T) = \frac{\gamma_d(1 - \beta T)}{2}$. Moreover, the second order condition is also satisfied since $\frac{\partial^2 \Pi}{\partial p_d^2} = -\frac{2\lambda_d}{\gamma_d(1 - \beta T)} < 0$. However, in this case, since γ_m is sufficiently high and $0 \leq T < \xi < \frac{1}{2\beta}$, we have

$$p_d^*(T) = \frac{\gamma_d(1 - \beta T)}{2} > \frac{p_m \gamma_d \delta(1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))}. \quad (42)$$

Hence, $\Pi(T, p_d)$ is increasing in p_d in this region of p_d .

Next, consider the second region of p_d , i.e., $\frac{p_m \gamma_d \delta(1 - \beta \xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta \xi))} < p_d \leq \frac{p_m \gamma_d(1 - \beta T)}{\gamma_m}$. In this region, following the similar argument as above, we obtain the consumer purchase equilibrium strategy for class L as follows:

$$\sigma^*(v, L) = \begin{cases} B & \text{if } \frac{p_d}{\delta \gamma_d(1 - \beta \xi)} \leq v \leq 1; \\ M & \text{if } \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)} \leq v < \frac{p_d}{\delta \gamma_d(1 - \beta \xi)}; \\ D & \text{if } \frac{p_d}{\gamma_d(1 - \beta T)} \leq v < \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)}; \\ N & \text{if } v < \frac{p_d}{\gamma_d(1 - \beta T)}. \end{cases} \quad (43)$$

In this region, the studio's profit function then becomes

$$\begin{aligned} \Pi(T, p_d) = & \lambda_m p_m \rho \left(1 - \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)} \right) + \\ & \lambda_d p_d \left(\rho \left(1 - \frac{p_d}{\delta \gamma_d(1 - \beta \xi)} + \frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)} - \frac{p_d}{\gamma_d(1 - \beta T)} \right) + (1 - \rho) \left(1 - \frac{p_d}{\gamma_d(1 - \beta T)} \right) \right). \end{aligned} \quad (44)$$

Solving the optimal video pricing problem given T by taking the derivative of $\Pi(T, p_d)$ in (44) with respect to p_d , we obtain

$$\frac{\partial \Pi}{\partial p_d} = \lambda_d + \frac{\rho p_m (\lambda_m + \lambda_d)}{\gamma_m - (1 - \beta T) \gamma_d} - \frac{2 \lambda_d p_d}{\gamma_d} \left(\frac{1}{1 - \beta T} + \frac{\rho (\gamma_m - \gamma_d (1 - \beta T - \delta (1 - \beta \xi)))}{\delta (1 - \beta \xi) (\gamma_m - (1 - \beta T) \gamma_d)} \right). \quad (45)$$

The second order condition is satisfied in this case:

$$\frac{\partial^2 \Pi}{\partial p_d^2} = - \frac{2 \lambda_d}{\gamma_d} \left(\frac{1}{1 - \beta T} + \frac{\rho (\gamma_m - \gamma_d (1 - \beta T - \delta (1 - \beta \xi)))}{\delta (1 - \beta \xi) (\gamma_m - (1 - \beta T) \gamma_d)} \right) < 0. \quad (46)$$

Equating the first order condition to zero, we have

$$p_d = \frac{\delta \gamma_d (1 - \beta T) (1 - \beta \xi) (\rho p_m (\lambda_m + \lambda_d) + \lambda_d (\gamma_m - (1 - \beta T) \gamma_d))}{2 \lambda_d ((1 - \beta \xi) (\gamma_m - (1 - \beta T) \gamma_d) \delta + (1 - \beta T) (\gamma_m - \gamma_d (1 - \beta T - \delta (1 - \beta \xi))) \rho)}, \quad (47)$$

which approaches $\frac{\delta \gamma_d (1 - \beta \xi) (1 - \beta T)}{2 (\rho (1 - \beta T) + \delta (1 - \beta \xi))}$, as γ_m gets large, since $0 < T < \xi < \frac{1}{2\beta}$. This value becomes larger than the upper bound of this corresponding region in p_d , which is $\frac{p_m \gamma_d (1 - \beta T)}{\gamma_m}$, as γ_m gets bigger. As a result, if γ_m is sufficiently big, the studio's profit function is increasing in p_d in this region of p_d , i.e., $\frac{p_m \gamma_d \delta (1 - \beta \xi)}{\gamma_m - \gamma_d (1 - \beta T - \delta (1 - \beta \xi))} < p_d \leq \frac{p_m \gamma_d (1 - \beta T)}{\gamma_m}$.

Next, if p_d increases further into the region of $\frac{p_m \gamma_d (1 - \beta T)}{\gamma_m} < p_d \leq \gamma_d \delta (1 - \beta \xi)$, again, following the similar argument above for the case of small p_d values, we obtain the consumer equilibrium market structure as given in (11). Then, the corresponding studio's profit function is written as

$$\Pi(T, p_d) = \lambda_m p_m \rho \left(1 - \frac{p_m}{\gamma_m} \right) + \lambda_d p_d \left(\rho \left(1 - \frac{p_d}{\delta \gamma_d (1 - \beta \xi)} \right) + (1 - \rho) \left(1 - \frac{p_d}{\gamma_d (1 - \beta T)} \right) \right). \quad (48)$$

Taking the derivative of this profit function $\Pi(T, p_d)$ in (48) with respect to p_d , we then obtain

$$\frac{\partial \Pi}{\partial p_d} = \frac{\delta \lambda_d (1 - \beta \xi) (\gamma_d (1 - \beta T) - 2 p_d) - 2 \rho \lambda_d p_d (1 - \beta T - \delta (1 - \beta \xi))}{\delta \gamma_d (1 - \beta T) (1 - \beta \xi)}. \quad (49)$$

Furthermore, the second order condition becomes

$$\frac{\partial^2 \Pi}{\partial p_d^2} = - \frac{2 \lambda_d}{\gamma_d} \left(\frac{1 - \rho}{1 - \beta T} + \frac{\rho}{\delta (1 - \beta \xi)} \right) < 0, \quad (50)$$

which is satisfied, and hence the first order condition is sufficient. Thus, $p_d(T)$ is an interior solution given T , i.e., given T , $p_d(T) = \frac{\delta \gamma_d (1 - \beta T) (1 - \beta \xi)}{2 (\delta (1 - \beta \xi) (1 - \rho) + \rho (1 - \beta T))}$. Furthermore, as γ_m is sufficiently high and $0 < T < \xi < \frac{1}{2\beta}$, it follows that $p_d(T) \in \left(\frac{p_m \gamma_d (1 - \beta T)}{\gamma_m}, \gamma_d \delta (1 - \beta \xi) \right)$. Hence, within this region, we achieve the local interior optimizer video price given T .

Lastly, if p_d increases further, i.e., $p_d \geq \gamma_d \delta (1 - \beta \xi)$, following the previous arguments, using the algebra and simplifying, we obtain the consumer equilibrium market structure as

$$\sigma^*(v, L) = \begin{cases} M & \text{if } \frac{p_m}{\gamma_m} \leq v < 1; \\ N & \text{if } v < \frac{p_m}{\gamma_m}. \end{cases} \quad (51)$$

In this region, the studio's profit function then becomes

$$\Pi(T, p_d) = \lambda_m p_m \rho \left(1 - \frac{p_m}{\gamma_m} \right) + \lambda_d p_d (1 - \rho) \left(1 - \frac{p_d}{\gamma_d (1 - \beta T)} \right). \quad (52)$$

Optimizing over p_d given T , we obtain the interior optimizer as $p_d(T) = \frac{\gamma_d (1 - \beta T)}{2}$, if it exists in the corresponding region. However, if δ is sufficiently high and $0 < T < \xi < \frac{1}{2\beta}$, then it becomes $p_d(T) = \frac{\gamma_d (1 - \beta T)}{2} < \gamma_d \delta (1 - \beta \xi)$. Consequently, the studio's profit function in (52) is decreasing in p_d given a fixed video release time T .

In summary, given T , the optimal video pricing problem has the optimizer as $p_d^*(T) = \frac{\delta \gamma_d (1 - \beta T) (1 - \beta \xi)}{2(\delta(1 - \beta \xi)(1 - \rho) + \rho(1 - \beta T))}$. Substituting this optimal video price $p_d^*(T)$ into the corresponding studio's profit function in (48), the studio's profit function in terms of T is written as

$$\Pi(T, p_d^*(T)) = \frac{\rho \lambda_m p_m (\gamma_m - p_m)}{\gamma_m} + \frac{\delta \lambda_d \gamma_d (1 - \beta T) (1 - \beta \xi)}{4(\delta(1 - \rho)(1 - \beta \xi) + \rho(1 - \beta T))}. \quad (53)$$

Taking the derivative of this profit function with respect to T , we have

$$\frac{d\Pi(T, p_d^*(T))}{dT} = -\frac{\beta \delta^2 \lambda_d \gamma_d (1 - \beta \xi)^2 (1 - \rho)}{4(\delta(1 - \beta \xi)(1 - \rho) + \rho(1 - \beta T))^2} < 0. \quad (54)$$

Hence, $T^* = 0 < \xi$, and plugging this back into $p_d^*(T^* = 0)$, we then obtain the optimal p_d^* .

So far, we have considered the case in which $T < \xi$. If $T \geq \xi$, following similar steps, we can show that under the conditions given in the Proposition, in particular, under high δ and γ_m , the consumer market structure becomes

$$\begin{cases} (none, video, both) & \text{if } 0 \leq p_d \leq \frac{p_m \gamma_d \delta (1 - \beta T)}{\gamma_m - \gamma_d (1 - \beta T) (1 - \delta)}; \\ (none, video, movie, both) & \text{if } \frac{p_m \gamma_d \delta (1 - \beta T)}{\gamma_m - \gamma_d (1 - \beta T) (1 - \delta)} < p_d \leq \frac{p_m \gamma_d (1 - \beta T)}{\gamma_m}; \\ (none, movie, both) & \text{if } \frac{p_m \gamma_d (1 - \beta T)}{\gamma_m} < p_d \leq \gamma_d \delta (1 - \beta T); \\ (none, movie) & \text{if } \gamma_d \delta (1 - \beta T) < p_d. \end{cases} \quad (55)$$

Using the similar algebra in the previous case of $T < \xi$, we find that in this region of T , the studio's profit $\Pi(T, p_d^*(T))$ is decreasing in T under the conditions given in this Proposition. As a result, the day-and-date strategy ($T^* = 0$ with both the movie and the video demands) is the optimal video release strategy in this region. ■

Proof of Proposition 4 Technically, we will prove that there exists $\omega_3 > 0$ such that if $\frac{\gamma_m}{\omega_3} > \omega_3$, $\alpha > \omega_3$ and $\delta < \frac{1}{\omega_3}$, i.e., γ_m and α are sufficiently high, and δ is sufficiently low, and $\rho \leq \frac{(1-2\delta)(1-\beta\xi)}{2(1-\delta(1-\beta\xi))}$, then $T^* = 0$ and $p_d^* = \frac{\gamma_d}{2}$. In addition, under this optimal strategy, the studio achieves perfect segmentation of consumers, i.e., $\sigma^*(v, L) \in \{N, M\}$ for all $v \in \mathcal{V}$ with $\sigma^*(v, L) = M$ with some mass of $v \in \mathcal{V}$, and $\sigma^*(v, H) \in \{N, D\}$ for all $v \in \mathcal{V}$ with $\sigma^*(v, H) = D$ with some mass of $v \in \mathcal{V}$.

First, we consider the case in which $T > \xi$. Under the given parameter region, in particular, for sufficiently high γ_m and α , and sufficiently low δ , the relevant consumer market structure becomes (51) for class L consumers, and (10) for class H consumers. Depending on the video price p_d and the video release time T , some other consumer market structures can also arise in equilibrium. However, as γ_m and α becomes sufficiently high and δ becomes sufficiently small, the potential region of p_d that induces those consumer equilibrium market structures becomes arbitrarily small, and hence, those consumer market structures will be dominated by the given relevant consumer market structure. In this consumer market structure, the movie demand is $\eta = \rho(1 - \frac{p_m}{\gamma_m})$, and the studio's profit function can be written as

$$\Pi(T, p_d) = \rho\lambda_m p_m \left(1 - \frac{p_m}{\gamma_m}\right) + (1 - \rho)\lambda_d p_d \left(1 - \frac{p_d}{\gamma_d(1 - \beta T)}\right). \quad (56)$$

By taking the derivative of this profit function with respect to p_d , we obtain

$$\frac{\partial \Pi(T, p_d)}{\partial p_d} = \frac{(1 - \rho)\lambda_d(\gamma_d(1 - \beta T) - 2p_d)}{\gamma_d(1 - \beta T)}. \quad (57)$$

The second order condition becomes $\frac{\partial^2 \Pi(T, p_d)}{\partial p_d^2} = -\frac{2(1-\rho)\lambda_d}{\gamma_d(1-\beta T)} < 0$, which is satisfied. Solving the first order condition, we have $p_d^*(T) = \frac{\gamma_d(1-\beta T)}{2}$, which is in the interior of the relevant region under the conditions given in the Proposition including low ρ condition. Next, by substituting this optimal video price $p_d^*(T)$ into the studio's profit function in (56), we obtain

$$\Pi(T, p_d^*(T)) = \frac{(1 - \rho)\lambda_d \gamma_m \gamma_d (1 - \beta T) + 4\rho\lambda_m p_m (\gamma_m - p_m)}{4\gamma_m}, \quad (58)$$

which is decreasing in T . Hence, $T^* = 0$ and consequently, $p_d^* = \frac{\gamma_d}{2}$.

Second, consider the case in which $T \geq \xi$. In this case, similar to the previous case of $T < \xi$, under the conditions given in the Proposition, the only relevant region is (51) for class L consumers, and (10) for class H consumers. In this consumer market equilibrium, from class L consumers, there is only *movie* demand, and the studio's profit from this class is $\rho\lambda_m p_m \left(1 - \frac{p_m}{\gamma_m}\right)$ as given in (56). However, this profit from class L is independent of p_d and T . In contrast, the studio's profit from class H is $(1 - \rho)\lambda_d p_d \left(1 - \frac{p_d}{\gamma_d(1 - \beta T)}\right)$, all from *video*, and it is strictly decreasing in T for all p_d . Hence, the studio's profit is strictly decreasing in T in this region.

As a result, $T^* = 0 < \xi$ and $p_d^* = \frac{\gamma_d}{2}$. In addition, the strictly positive *movie* demand is from class L and strictly positive *video* demand from the class H without *both* segment from neither class. ■

Proof of Proposition 5 Technically, we will prove that there exist $\omega_4 > 0$ such that for $\alpha > \omega_4$ and $\delta > 1 - \frac{1}{\omega_4}$, when $\frac{\gamma_m}{p_m} < \frac{\gamma_m}{p_m} < \overline{\gamma_m}$, $\lambda_d < \lambda_m$ and $\gamma_d > \underline{\gamma_d}$ are satisfied, where $\underline{\gamma_m} = \sqrt{\lambda_m \rho / \lambda_d}$, $\overline{\gamma_m} = \min \left(\frac{1}{2p_m} \left(\gamma_d(1 - \beta\xi) + \sqrt{\gamma_d^2(1 - \beta\xi)^2 + \delta p_m^2} \right), 2(1 - \beta\xi), 2(1 - \rho) \right)$, and $\underline{\gamma_d} = \frac{p_m((\lambda_m - \lambda_d)\rho - 2\lambda_d(1 - \beta\xi))}{\lambda_d}$, then the following holds:

(i) $\gamma_m/p_m < 2\sqrt{\lambda_m \rho / \lambda_d}$, then $T^* = \xi$ and $p_d^* = \gamma_d(1 - \beta\xi)/2$.

(ii) Otherwise, $T^* = \frac{1 - \delta(1 - \beta\xi)}{\beta} - \frac{\gamma_m}{\beta\gamma_d} + \frac{2p_m}{\beta\gamma_d} \sqrt{\frac{\lambda_m \rho}{\lambda_d}}$ and $p_d^* = \frac{\gamma_m + \delta\gamma_d(1 - \beta\xi)}{2} - p_m \sqrt{\frac{\lambda_m \rho}{\lambda_d}}$.

We first consider the region in which $T < \xi$. Under the conditions given in the Proposition, the consumer equilibrium market structure is presented in (38) as p_d changes given a fixed $T > \xi$. In this region, using the conditions from the Proposition, we can show that the profit function is decreasing in p_d if $p_d > \frac{p_m \gamma_d \delta(1 - \beta\xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta\xi))}$. When $p_d \leq \frac{p_m \gamma_d \delta(1 - \beta\xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta\xi))}$, the optimal video price $p_d^*(T)$ becomes an interior optimal solution; that is, given T , $p_d^*(T) = \frac{\gamma_d(1 - \beta T)}{2} < \frac{p_m \gamma_d \delta(1 - \beta\xi)}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta\xi))}$ under the conditions given in the Proposition. Substituting $p_d^*(T)$ into the profit function in (40) and simplifying, we obtain

$$\Pi(T, p_d^*(T)) = \frac{\lambda_d \gamma_d(1 - \beta T)}{4} + \rho \lambda_m p_m \left(1 - \frac{p_m}{\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta\xi))} \right). \quad (59)$$

Taking the derivative of $\Pi(T, p_d^*(T))$ with respect to T , we then have

$$\frac{d\Pi(T, p_d^*(T))}{dT} = \beta \gamma_d \left(-\frac{\lambda_d}{4} + \frac{\rho \lambda_m p_m^2}{(\gamma_m - \gamma_d(1 - \beta T - \delta(1 - \beta\xi)))^2} \right). \quad (60)$$

Note that $\frac{d\Pi(T, p_d^*(T))}{dT}$ is decreasing in T , and hence the second order condition is satisfied. Furthermore, (60) is strictly positive for all $T < \xi$ under sufficient high δ , if $\frac{\gamma_m}{p_m} < 2\sqrt{\frac{\lambda_m \rho}{\lambda_d}}$. As a result, in this case, $T^* = \xi$. Plugging $T^* = \xi$ into $p_d^*(T)$, we obtain $p_d^*(\xi) = \frac{\gamma_d(1 - \beta\xi)}{2}$. Otherwise, i.e., if $\frac{\gamma_m}{p_m} > 2\sqrt{\frac{\lambda_m \rho}{\lambda_d}}$, the first order condition is satisfied at $T^* = \frac{1 - \delta(1 - \beta\xi)}{\beta} - \frac{\gamma_m}{\beta\gamma_d} + \frac{2p_m}{\beta\gamma_d} \sqrt{\frac{\lambda_m \rho}{\lambda_d}}$ and by replacing this optimal T^* , we then obtain p_d^* . Lastly, when $T > \xi$, similarly, it follows that the studio's profit is decreasing in T under the parameter region of the Proposition. Hence, it completes the proof. ■

Proof of Proposition 6 Technically, we will prove that there exists $\omega_5 > 0$ such that for $\frac{\gamma_m}{p_m} > \omega_5$, $\alpha < \frac{1}{\omega_5}$, the following holds:

(i) If $\delta > 1 - \frac{1}{\omega_5}$, then $T^* = \xi$ and $p_d^* = \frac{\delta\gamma_d(1 - \beta\xi)}{2}$;

(ii) If $\delta < \frac{1}{\omega_5}$ and $\lambda_m < \lambda_d$, $T^* = 0$ and $p_d^* = \frac{\gamma_d(p_m(\lambda_m + \lambda_d) + \alpha\lambda_d(1 - \rho))}{2\lambda_d(\gamma_m + \alpha(1 - \rho))}$.

For part (i), we first consider $T \leq \xi$, and in this case, the relevant equilibrium market structure is given in (11) for class L . The equilibrium consumer market structure for Class H has also the same

structure as below, but with different thresholds:

$$\sigma^*(v, H) = \begin{cases} B & \text{if } \frac{p_d}{\delta\gamma_d(1-\beta\xi)} \leq v < 1; \\ M & \text{if } \frac{\gamma_m(p_m + \alpha(1-\rho)) + \alpha\rho(\gamma_m - p_m)}{\gamma_m(\gamma_m + \alpha(1-\rho))} \leq v < \frac{p_d}{\delta\gamma_d(1-\beta\xi)}; \\ N & \text{if } v < \frac{\gamma_m(p_m + \alpha(1-\rho)) + \alpha\rho(\gamma_m - p_m)}{\gamma_m(\gamma_m + \alpha(1-\rho))}. \end{cases} \quad (61)$$

In this region, the studio's profit function is written as

$$\Pi(T, p_d) = \frac{\lambda_m p_m (\gamma_m - p_m)}{\gamma_m + \alpha(1 - \rho)} + \lambda_d p_d \left(1 - \frac{p_d}{\delta\gamma_d(1 - \beta\xi)} \right). \quad (62)$$

First, this profit function does not depend on T . Since the actual consumption for consumers occur at ξ for *both* consumers, we resolve this multiple optimizers to be $T^* = \xi$. Higher T^* is also more likely due to the production lead time for the videos. Moreover, as α gets bigger, T^* also increases starting from ξ in order to induce *both* segment. So, in this case of low α , $T^* = \xi$, and then by maximizing the studio's profit function over p_d , we obtain $p_d^* = \frac{\delta\gamma_d(1-\beta\xi)}{2}$. Similarly, we can show that if $T > \xi$, the studio's profit is decreasing in T in this parameter region. Hence, under high content durability, $T^* = \xi$ and $p_d^* = \frac{\delta\gamma_d(1-\beta\xi)}{2}$.

For part (ii), the relevant equilibrium consumer market structures are (14) for class H and (14) with $\alpha = 0$ for class L . Then, the corresponding theatrical movie demand is $D_m = \eta = \frac{\gamma_m - p_m - (\gamma_d(1-\beta T) - p_d)}{\gamma_m - \gamma_d(1-\beta T) + \alpha(1-\rho)}$. In addition, the video demand is

$$D_d = \rho \left(\frac{p_m - p_d}{\gamma_m - \gamma_d(1 - \beta T)} - \frac{p_d}{\gamma_d(1 - \beta T)} \right) + (1 - \rho) \left(\frac{1}{\gamma_m - \gamma_d(1 - \beta T)} \left(p_m - p_d + \frac{\alpha(\gamma_m - p_m - (\gamma_d(1 - \beta T) - p_d))}{\gamma_m - (1 - \beta T)\gamma_d + \alpha(1 - \rho)} \right) - \frac{p_d}{(1 - \beta T)\gamma_d} \right). \quad (63)$$

Then, the studio's profit function is written as $\Pi(T, p_d) = \lambda_m p_m D_m + \lambda_d p_d D_d$, which is a concave function in p_d for a given T . Optimizing $\Pi(T, p_d)$ over p_d given T , we obtain

$$p_d^*(T) = \frac{\gamma_d(1 - \beta T)((-d + \lambda_m)p_m + \alpha\lambda_d(1 - \rho))}{2\lambda_d(\gamma_m + \alpha(1 - \rho))}. \quad (64)$$

Substituting the optimal video price $p_d^*(T)$ into the studio's profit function, we obtain $\Pi(T, p_d^*(T))$.

Taking the derivative of this profit function with respect to T , we have

$$\frac{d\Pi(T, p_d^*(T))}{dT} = -\frac{\beta\gamma_d(p_m(\lambda_d - \lambda_m) + \alpha\lambda_d(1 - \rho))^2}{4\lambda_d(\gamma_m - (1 - \beta T)\gamma_d + \alpha(1 - \rho))^2} < 0. \quad (65)$$

As a result, the optimal video release time in this region is $T^* = 0$. Replacing $T^* = 0$ and $p_d^*(0)$ into the studio's profit function, we obtain the optimal profit function in this consumer market structure. Lastly, if $\lambda_d > \lambda_m$, the equilibrium market structure that we considered here dominates the optimal studio's profit under other equilibrium consumer market structures. Therefore, in this

parameter region, $T^* = 0$ and $p_d^* = \frac{\gamma_d(p_m(\lambda_m + \lambda_d) + \alpha\lambda_d(1-\rho))}{2\lambda_d(\gamma_m + \alpha(1-\rho))}$. ■

Proof of Proposition 7 Technically, we will prove that there exists $\omega_6 > 0$ such that for $\frac{\gamma_m}{p_m} > \omega_6$, and $\delta < \frac{1}{\omega_6}$, if $\lambda_d > \lambda_m \max(1, \frac{4p_m}{\gamma_d})$, the following holds

$$\Pi^*|_{\{\alpha < 1/\omega_6\}} < \Pi^*|_{\{\alpha > \omega_6\}}. \quad (66)$$

If α is sufficiently high, it corresponds to the perfect segmentation in Proposition 4, whereas if α is sufficiently low, it corresponds to the delayed release in part (ii) of Proposition 5 under the condition of $\lambda_d > \lambda_m$. By comparing the optimal studio's profit in those two cases, we obtain that if $\lambda_d > \frac{4\lambda_m p_m}{\gamma_d}$, the studio's profit under perfect segmentation is higher than the profit under delayed release. ■

Proof of Proposition 8 Technically, we will prove that there exists $\omega_7 > 0$ such that for $\frac{\gamma_m}{p_m} > \omega_7$, $\alpha > \omega_7$, and $\delta < \frac{1}{\omega_6}$, then the following holds:

(i) If $\rho \leq \frac{(1-2\delta)(1-\beta\xi)}{2(1-\delta(1-\beta\xi))}$, then $T^* = 0$ and $p_d^* = \frac{\gamma_d}{2}$.

(ii) If $\rho > 1 - \frac{1}{\omega_7}$, then $T^* = \xi$ and $p_d^* = \frac{p_m \gamma_d (1-\beta\xi)}{\gamma_m}$.

Part (i) is from the proof of Proposition 4. Part (ii) is similar to the proof of Proposition 3. The difference is that instead of sufficiently high δ , we now have sufficiently high ρ and sufficiently low δ . Closely following the proof of Proposition 3, we first prove that $p_d^*(T) = \frac{p_m \gamma_d (1-\beta T)}{\gamma_m}$ when $T \leq \xi$. Moreover, in this region of $T < \xi$, the studio's profit is increasing in T under high ρ . In addition, if $T \geq \xi$, the profit is decreasing in T , which leads to $T^* = \xi$. ■

Proof of Proposition 9 Technically, we will prove that there exists $\omega_8 > 0$ such that for $\alpha > \omega_8$ and $\delta < \frac{1}{\omega_9}$, if there exists a low video price bound c_d , and $\lambda_d > \rho\lambda_m$ and

$$\max\left(1 + \frac{\gamma_d}{2p_m}, \frac{2\sqrt{\lambda_d \lambda_m \rho} - (\lambda_d + \lambda_m)\rho}{\lambda_d(1-\rho)}\right) < \frac{\gamma_m}{p_m} < \min\left(\frac{1}{\delta(1-\beta\xi)}, \frac{2\lambda_d - (\lambda_d + \lambda_m)\rho}{\lambda_d(1-\rho)}, 2\right), \quad (67)$$

are satisfied, then the following holds:

(i) If $\gamma_d = K_1 c_d$, for some $K_1 > 1$, i.e., if γ_d is also low, then $p_d^* = \frac{\gamma_d(\lambda_d(1-\rho)(\gamma_m - \gamma_d) + \rho p_m(\lambda_m + \lambda_d))}{2\lambda_d(\gamma_m - (1-\rho)\gamma_d)} < c_d$ and $T^* = 0$;

(ii) If $K_2 < \gamma_d < \gamma_m$, for some $K_2 > 0$, then $p_d^* = c_d$ and $T^* > 0$ and satisfies:

$$T^* = \inf\left\{T \mid \rho\gamma_d^2(p_m - c_d)(\lambda_m p_m - \lambda_d c_d)(1 - \beta T)^2 - \lambda_d c_d^2(\gamma_m - \gamma_d(1 - \beta T))^2 = 0\right\}. \quad (68)$$

Moreover, when c_d becomes sufficiently small, T^* approaches $\frac{1}{\beta} \left(1 - \frac{\gamma_m c_d}{p_m \gamma_d} \sqrt{\frac{\lambda_d}{\rho \lambda_m}}\right)$.

In this case, the relevant equilibrium consumer market structure is (14) with $\alpha = 0$ for class L as studied in part (ii) of Proposition 6. For class H , it is (10) given high α as proved in Lemma 1. Closely following the proof of part (ii) of Proposition 6, using the conditions in the Proposition, and checking the boundary conditions including the video pricing boundary, the results follow. ■

Proof of Proposition 10 Technically, we will prove that in addition to the conditions in Proposition 9, if $\gamma_m \leq \frac{p_m \left((\lambda_m - \lambda_d)\rho + 2\sqrt{\rho(\lambda_m\rho(\lambda_m - 2\lambda_d) + \lambda_d^2(3 - \rho - \rho^2)) / (1 - \rho)} \right)}{\lambda_d(3 + \rho)}$, low ρ and $3\lambda_m < \lambda_d$ are satisfied, then

$$SW \big|_{\{\gamma_d = K_1 c_d\}} > SW \big|_{\{K_2 < \gamma_d < \gamma_m\}} . \quad (69)$$

Using the equilibrium outcome and consumer market structure as well as the welfare expressions in (9), we obtain the social welfare for both small γ_d and large γ_d case. Using the conditions given in this Proposition with some algebra, the result follows. The exact expressions are available from the authors upon request. ■